# Prototypes, Patents and Intellectual Property

Lecture 12

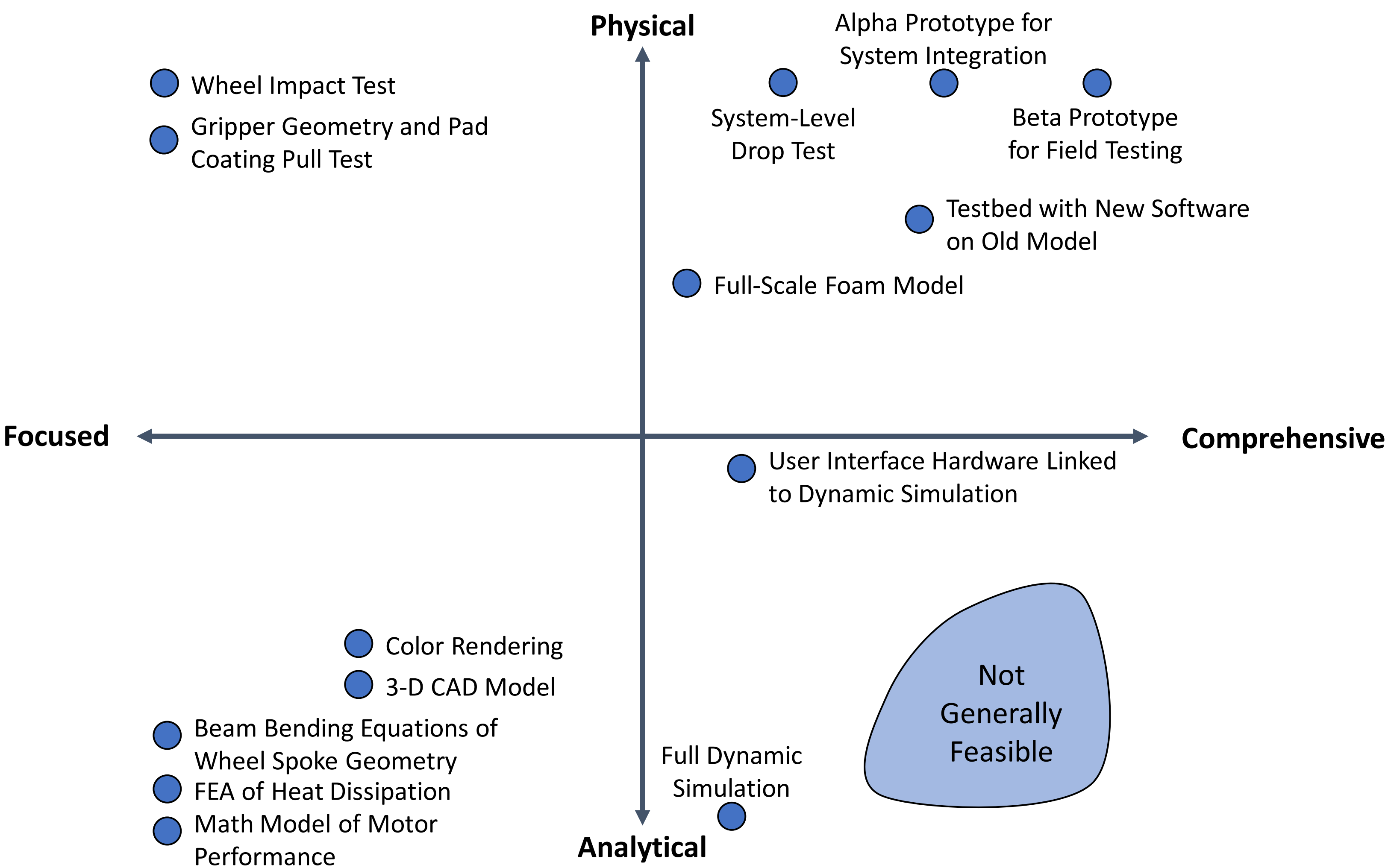
# Agenda

- Prototyping
- Tips on Prototyping using 3D printing
- Intellectual Property
- Engineering Idea #5: Small World Networks

# Four Uses of Prototypes

- Learning
  - answering questions about performance or feasibility
  - e.g., proof-of-concept model
- Communication
  - demonstration of product for feedback
  - e.g., 3D physical models of style or function
- Integration
  - combination of sub-systems into system model
  - e.g., alpha or beta test models
- Milestones
  - goal for development team's schedule
  - e.g., first testable hardware

# Types of Prototypes




Ulrich and Eppinger (4th ed., 2008) Exhibit 12-5

# Physical vs. Analytical Prototypes

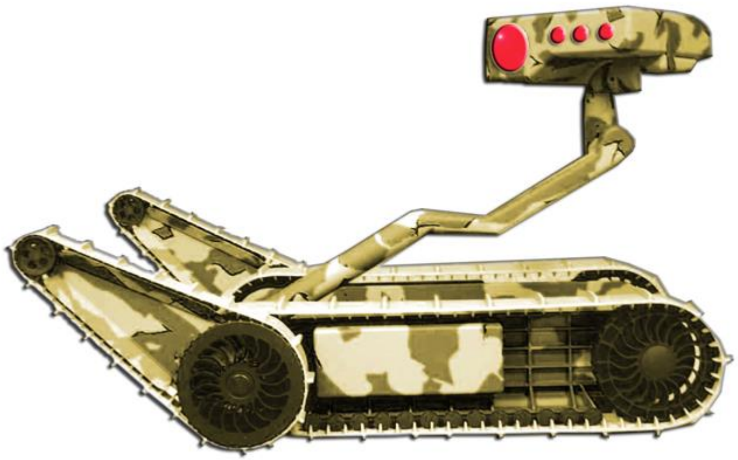
Physical Prototypes

- Tangible approximation of the product.
- May exhibit unmodeled behavior.
- Some behavior may be an artifact of the approximation.
- Often best for communication.

Analytical Prototypes

- Mathematical model of the product.
- Can only exhibit behavior arising from explicitly modeled phenomena. (However, behavior is not always anticipated).
- Some behavior may be an artifact of the analytical method.
- Often allow more experimental freedom than physical models.

# Physical Prototypes



Looks-like model for customer communication and approval

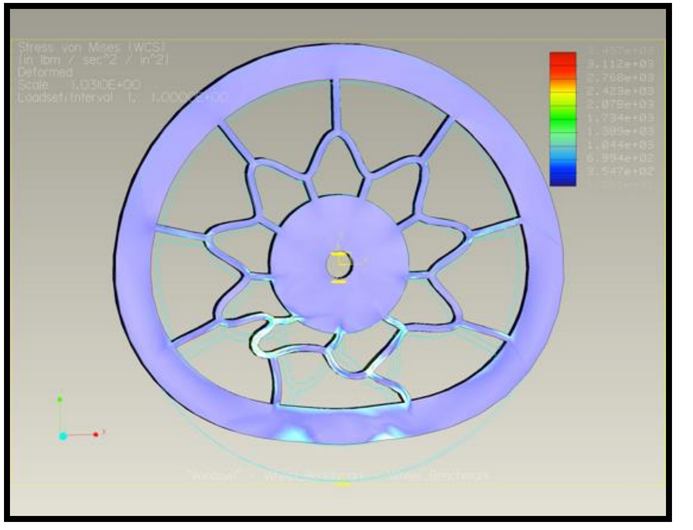
Wheel prototype under load during creep testing

Sand test

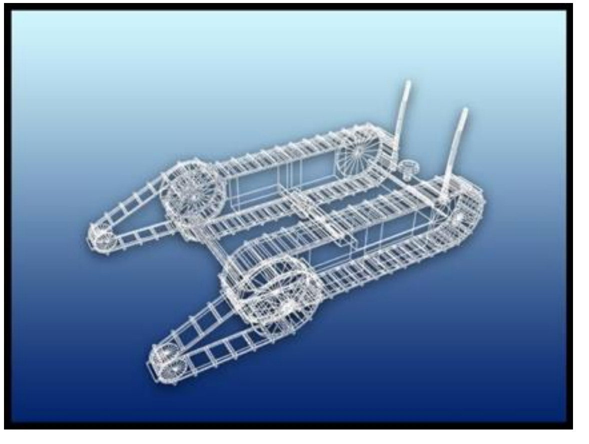
**Ulrich and Eppinger (4th ed., 2008) Exhibit 12-3**

# Analytical Prototypes

3D CAD rendering




Finite-element analysis



Dynamic simulation model

**Ulrich and Eppinger (4th ed., 2008) Exhibit 12-4**

# Focused vs. Comprehensive Prototypes

**Focused Prototypes**

- Implement one or a few attributes of the product.
- Answer specific questions about the product design.
- Generally several are required.

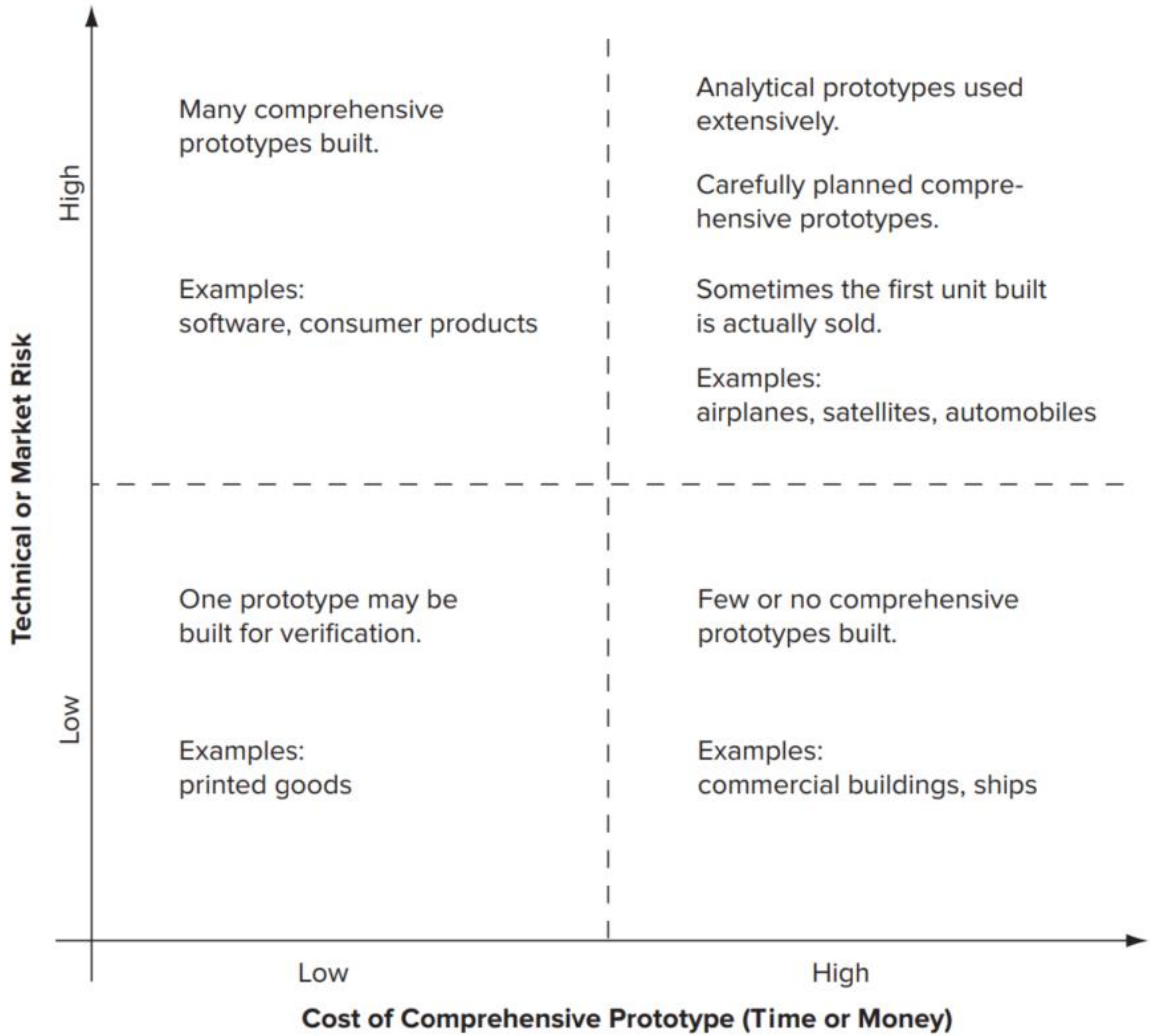
**Comprehensive Prototypes**

- Implement many or all attributes of the product.
- Offer opportunities for rigorous testing.
- Often best for milestones and integration.

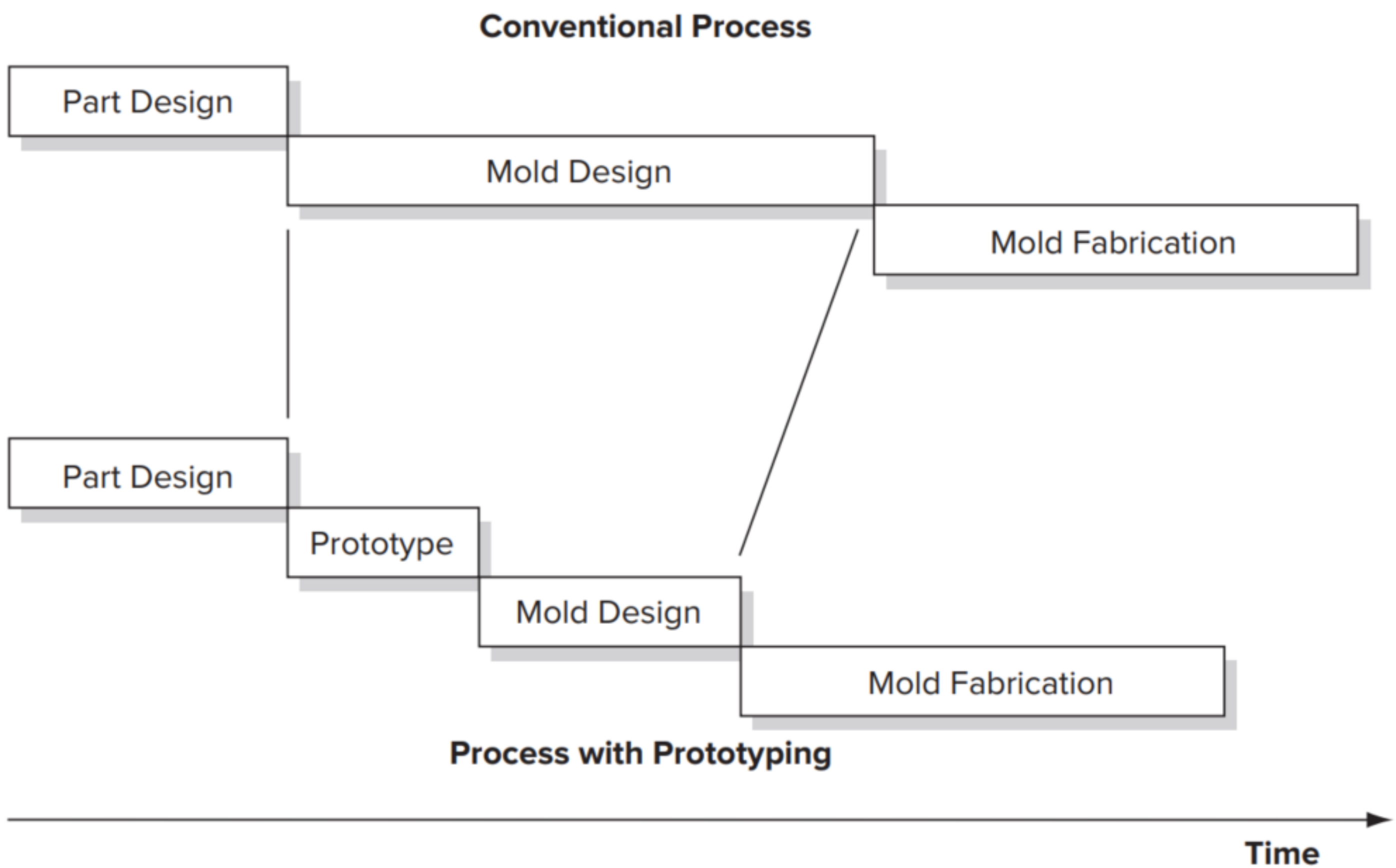
Wheel impact test

Beta prototype

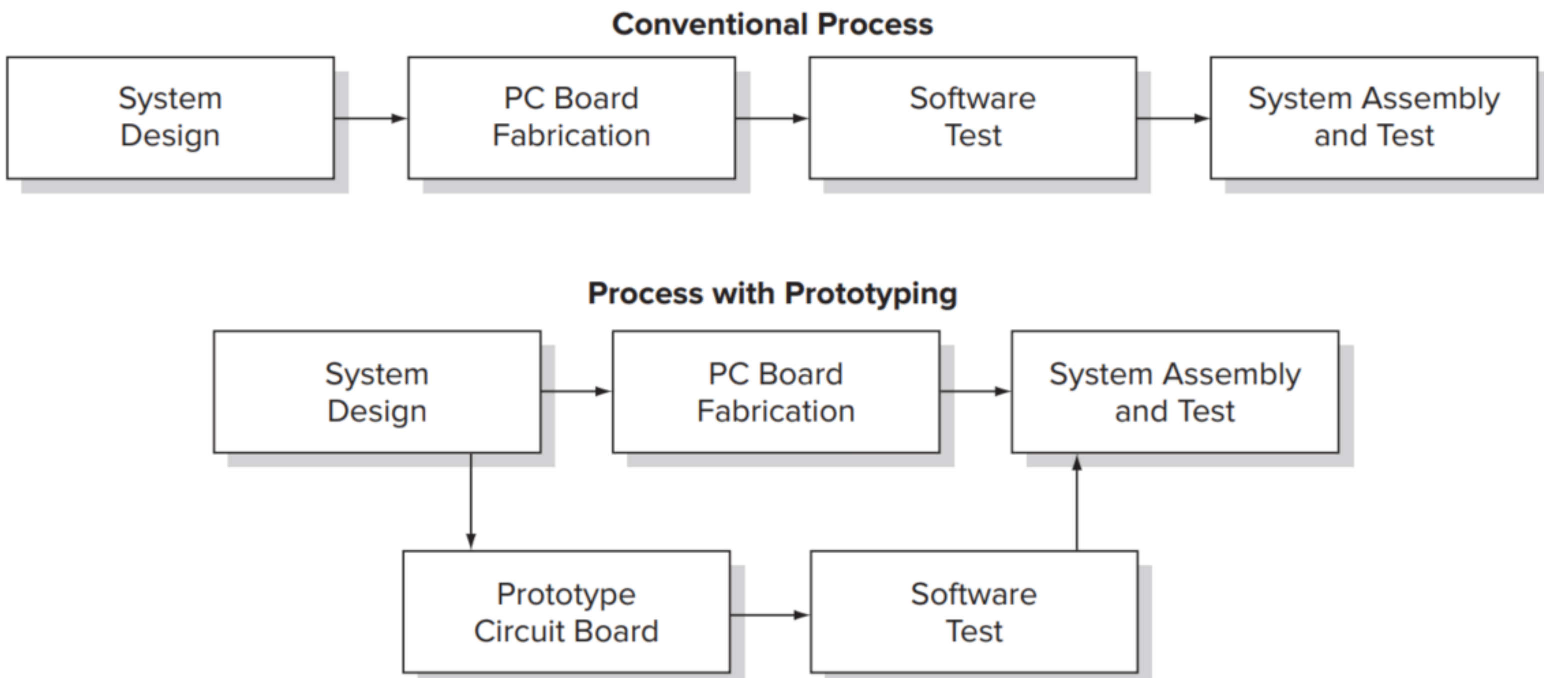
# How many comprehensive prototypes are necessary?



# Prototyping saves time and money while designing molds



# Prototyping saves time and money while testing software



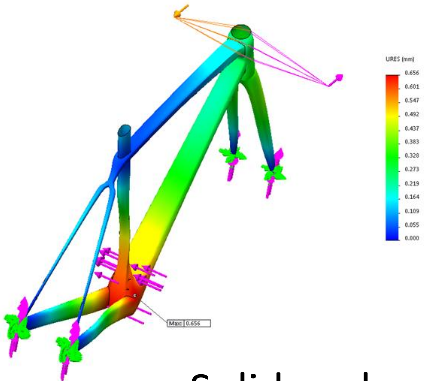
# Prototyping Strategy

- Use prototypes to reduce uncertainty
- Make models with a defined purpose
- Consider multiple forms of prototypes
- Choose the timing of prototype cycles
  - Many early models are used to validate concepts.
  - Relatively few comprehensive models are necessary to test integration.
- Plan time to learn from prototype cycles.
  - Avoid the "hardware swamp"

# Virtual Prototyping

3D CAD models enable many types of analysis:

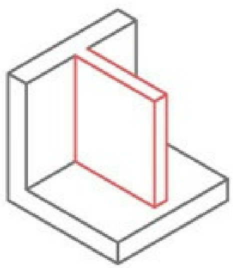
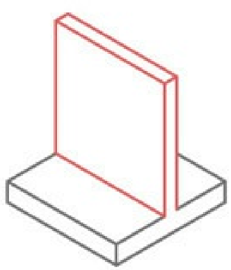
- Fit and assembly
- Manufacturability
- Form and style
- Kinematics
- Finite element analysis (stress, thermal)
- Crash testing
- better every year...



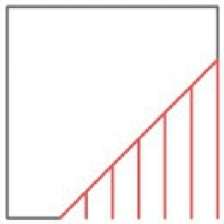
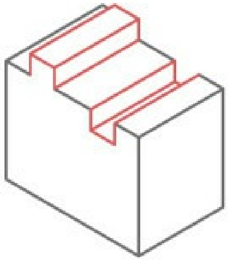
Solidworks

ESI Group

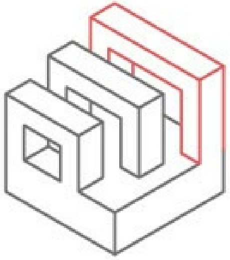
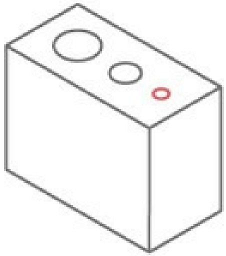
# Design Considerations for 3D Printing

| | |
|---|---|
| Supported walls  | Minimum thickness for walls that are connected to other structures on at least two sides, so they have little chance of warping. |
| Unsupported walls  | Minimum thickness for walls that are connected to the rest of the print on just one side, with a higher chance for warping or detaching from the print. |

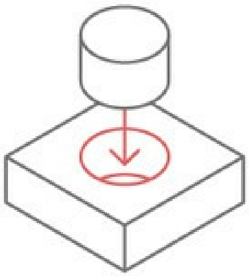
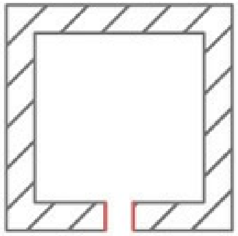
# Design Considerations for 3D Printing

| | |
|---|---|
| Overhangs  | The minimum angle (relative to the horizontal) a wall can be printed out without requiring support. |
| Embossed and engraved details  | The minimum feature depth / height (including text) which are imprinted or recessed into the model. These details are at risk of fusing with the rest of the model while printing if they are too small. |

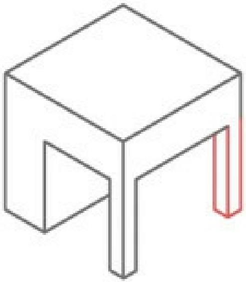
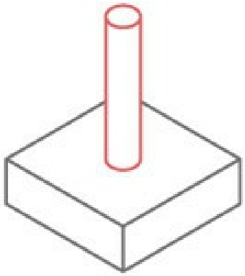
# Design Considerations for 3D Printing

| | |
|---|---|
| Horizontal bridges  | The maximum bridge length between two points on a model that can be successfully printed without the need for support material. |
| Holes  | The minimum diameter a technology can successfully print a hole. |

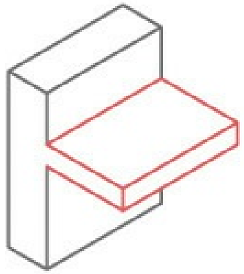
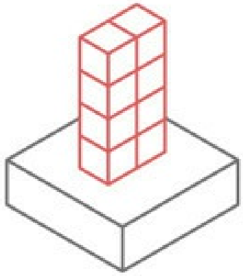
# Design Considerations for 3D Printing

| | |
|---|---|
| Connecting/moving parts  | The recommended clearance based on the required fit.If no fit is specified, the connection is assumed to be an interlocking fit. |
| Escape holes  | The minimum size escape holes must be to allow for the removal of the build material. To save weight (and sometimes costs) parts can be printed hollow. To remove build material after production escape holes must be included. |

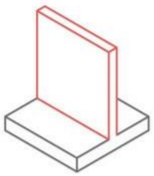
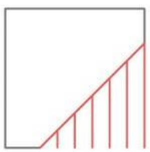
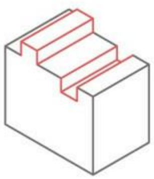
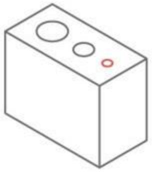
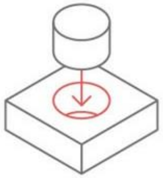
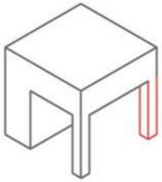
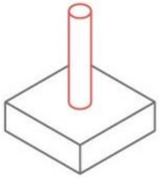
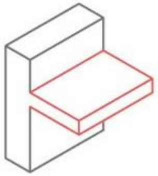
# Design Considerations for 3D Printing

| | |
|---|---|
| Minimum feature  | The minimum thickness of any feature is required to be to ensure it will successfully print. |
| Pin diameter  | The minimum reliable diameter a pin can be printed at. |

# Design Considerations for 3D Printing

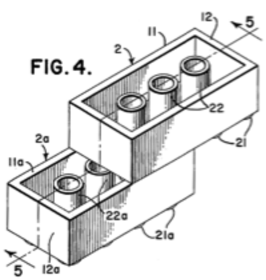
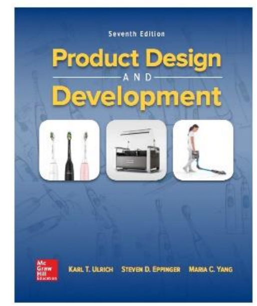
| | |
|---|---|
| Unsupported Edges  | The maximum length of a cantilever-style overhang. |
| Aspect Ratio  | The maximum ratio between the vertical print height and the part cross section to ensure stability of the printed part on the build plate. |

# FFF Printing Guidelines

| | |
|---|---|
| Wall thickness<br> | 0.8 mm<br><br>FFF is able to produce walls down to 0.8 mm. As a general design rule of thumb it is good to make wall thickness a multiple of the nozzle diameter (if this is known).The most common nozzle diameter for FFF printers is 0.4mm. |
| Overhangs<br> | 45°<br><br>Overhanging features less than 45° relative to the horizotal require support material to accurately print. |
| Embossed and engraved details<br> | 0.6 mm wide x 2 mm high<br><br>All embossed and engraved details produced via FFF should be no smaller than 0.6 mm wide and 2.0 mm high (if they are to be readable). |
| Bridges<br> | 10 mm<br><br>Unsupported horizontal bridges produced via FFF should have a span no greater than 10 mm to avoid sagging. |
| Holes<br> | Ø2.0 mm<br><br>FFF can sometimes produce holes that have a slightly undersize diameter. If accurate holes are required drilling after printing is recommended. Holes smaller than 2.0 mm in diameter should be avoided. |
| Clearance<br> | 0.5 mm<br><br>When clearance is required between parts, a spacing of 0.5 mm should be used. This can be varied based upon the type of fit that is required. |
| Feature size<br> | 2.0 mm<br><br>The minimum feature size when printing with FFF is 2.0 mm. |
| Pins<br> | Ø3.0 mm<br><br>Vertical pins created using FFF should be no smaller than 3.0 mm in diameter, if they are functional. Most of the issues relating to minimum pin size center around cooling. Off the shelf pins, inserted into a drilled hole, are the best solution if functional pins are required in a design. |
| Unsupported edges<br> | Ø3.0 mm<br><br>Unsupported edges will begin to lose quality and fail to print if they are too long. Unsupported edges should be no longer than 3.0 mm |

# Types of Intellectual Property

**Intellectual Property**

| Utility Patent | Design Patent | Plant Patent | Trademark | Copyright | Trade Secret | Geographical Indication |
|---|---|---|---|---|---|---|
|  |  |  | TM |  |  |  |
| 1. novel<br>2. useful<br>3. non-obvious | ornamental design only | new composition of matter | word or symbol | original expression of work | proprietary and useful | unique regional qualities |
| requires formal application | | | may be registered | | not registered | registered |

# Trademark

- Valuable for building and protecting a brand
- A “mark” under which you sell goods and services
  - House trademark
  - Product trademark ThinkPad
    - Trade dress Advil
    - Service mark
  - Federal registration available in US (TM or ®)
- Strength of trademark depends on distinction
  - Generic GIGABYTE™
  - Descriptive ANALOG DEVICES
  - Suggestive HARD AS NAILS®
  - Arbitrary iPad
  - Fanciful OXO

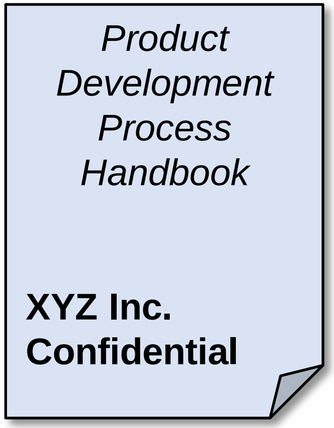
generally stronger

# Copyright

- The right to make copies
  - Arises from simply creating a work
  - Protects the *expression* – not the idea
    - great for music, poor for software
- Default copyright ownership
  - Owned by author unless otherwise agreed (e.g. by employee or contractor agreement)
- Open source
  - For sharing and building
- Federal registration is a plus
- © notice format is flexible
  - Copyright © 2018, MyBiz, Inc., All Rights Reserved
  - © 2018 MyBiz
  - Copyright MyBiz

# Trade Secret

- Confidential information that is used for competitive advantage
- Prevents, but does not block, others from developing similar knowledge
- Protection varies by state and country
- Lasts as long as you can keep it secret
- Must actively work to protect trade secrets
  - Nondisclosure agreements
  - Confidential markings
  - Employee education



# Patent

- **A limited-time monopoly, granted by government, in exchange for publicly sharing new, useful knowledge**
  - 20 years from filing date in US, Europe, and many other countries
  - **Periodic maintenance fees (**on utility patents, **not plant or design)**
- Gives owner the **right to exclude others from practicing their invention**
  - Owner's right to practice may be limited by others' prior patent rights (e.g. when extending a patent with more specific claims)
- **Real estate analogy:**
  - **Right to prevent trespassers**
  - **Ownership ≠ right to use –**
    **limited by access rights, zoning, etc.**
  - **Claims of patent ≈ fence around property**
- Patent search:
  - US www.u**spto.gov**
  - US **www.google.com**/patents
  - Global www.wipo.int

# Requirements to Obtain a Patent

- Patentable subject matter
- Not previously sold or publicly described
- **First to file**
- **Novel**
  - beyond what is already patented or known
  - prior art must be cited
- **Useful**
  - for some demonstrable need or value
  - initial commercial success may demonstrate
- **Not obvious**
  - “to one of ordinary skill in the art”
  - prior art “teaches against”
  - inventive, rather than simple modification

# Patent Application

**Patent application includes text and diagrams:**

- **Field of the invention**
  - Describe the problem addressed
- **Background of the invention**
  - Describe the prior art
  - List advantages over existing methods
- **Detailed description**
  - Best mode: the best way to implement the invention
  - **Examples of use and modes** of implementation
- **Claims**
  - **What <u>exactly</u> is the invention**

Provisional patent application (optional)

- **Establishes date of filing before examination begins**
  - 1 year to file full regular application
  - Public disclosure allowed after provisional is filed

# Chapter Example: David W. Coffin, Sr. US Pat.# 5,205,473 Coffee cup sleeve


US005205473A

**United States Patent** [19]
**Coffin, Sr.**

[11] **Patent Number: 5,205,473**
[45] **Date of Patent: Apr. 27, 1993**

[54] **RECYCLABLE CORRUGATED BEVERAGE CONTAINER AND HOLDER**

[75] Inventor: **David W. Coffin, Sr.**, Fayetteville, N.Y.

[73] Assignee: **Design By Us Company**, Philadelphia, Pa.

[21] Appl. No.: **854,425**

[22] Filed: **Mar. 19, 1992**

[51] **Int. Cl.**$^5$ .......... **B65D 3/28**
[52] **U.S. Cl.** .......... **229/1.5 B**; 206/813; 220/441; 220/DIG. 30; 229/1.5 H; 229/DIG. 2; 493/296; 493/907
[58] **Field of Search** .......... 229/1.5 B, 1.5 H, 4.5, 229/DIG. 2; 220/441, 671, 737–739, DIG. 30; 493/287, 296, 907, 908; 209/8, 47, 215; 206/813

[56] **References Cited**

U.S. PATENT DOCUMENTS

| | | | |
|---|---|---|---|
| 1,732,322 | 10/1929 | Wilson et al. | 220/DIG. 30 |
| 1,771,765 | 7/1930 | Benson | 229/4.5 |
| 2,266,828 | 12/1941 | Sykes | 229/1.5 B |
| 2,300,473 | 11/1942 | Winkle | 229/4.5 |
| 2,501,815 | 3/1950 | Hamm . | |
| 2,617,549 | 11/1952 | Egger . | |
| 2,641,402 | 6/1953 | Bruun | 229/4.5 |
| 2,661,889 | 12/1953 | Phinney | 229/4.5 |
| 2,969,901 | 1/1961 | Behrens | 229/1.5 B |
| 3,237,834 | 3/1966 | Davis et al. | 229/1.5 B |
| 3,779,157 | 12/1973 | Ross, Jr. et al. | 53/527 |
| 3,785,254 | 1/1974 | Mann . | |
| 3,890,762 | 6/1975 | Ernst et al. . | |
| 3,908,523 | 9/1975 | Shikaya | 229/1.5 B |
| 4,080,880 | 3/1978 | Shikay | 493/296 |
| 4,146,660 | 3/1979 | Hall et al. . | |
| 4,176,054 | 11/1979 | Kelley | 209/8 |
| 5,009,326 | 4/1991 | Reaves et al. . | |
| 5,092,485 | 3/1992 | Lee | 229/1.5 B |

OTHER PUBLICATIONS

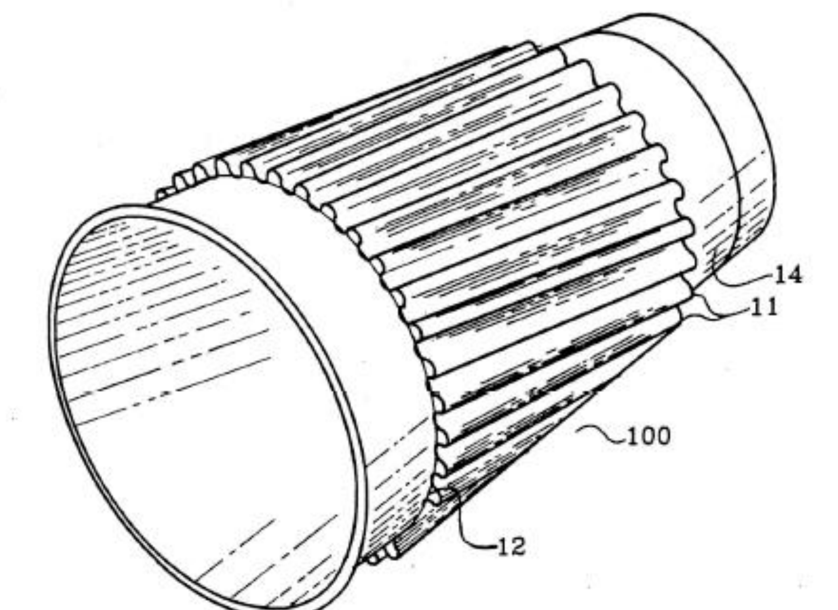
"The Wiley Encyclopedia of Packaging Technology", John Wiley & Sons, pp. 66–69, 1986.

*Primary Examiner*—Gary E. Elkins
*Attorney, Agent, or Firm*—Synnestvedt & Lechner

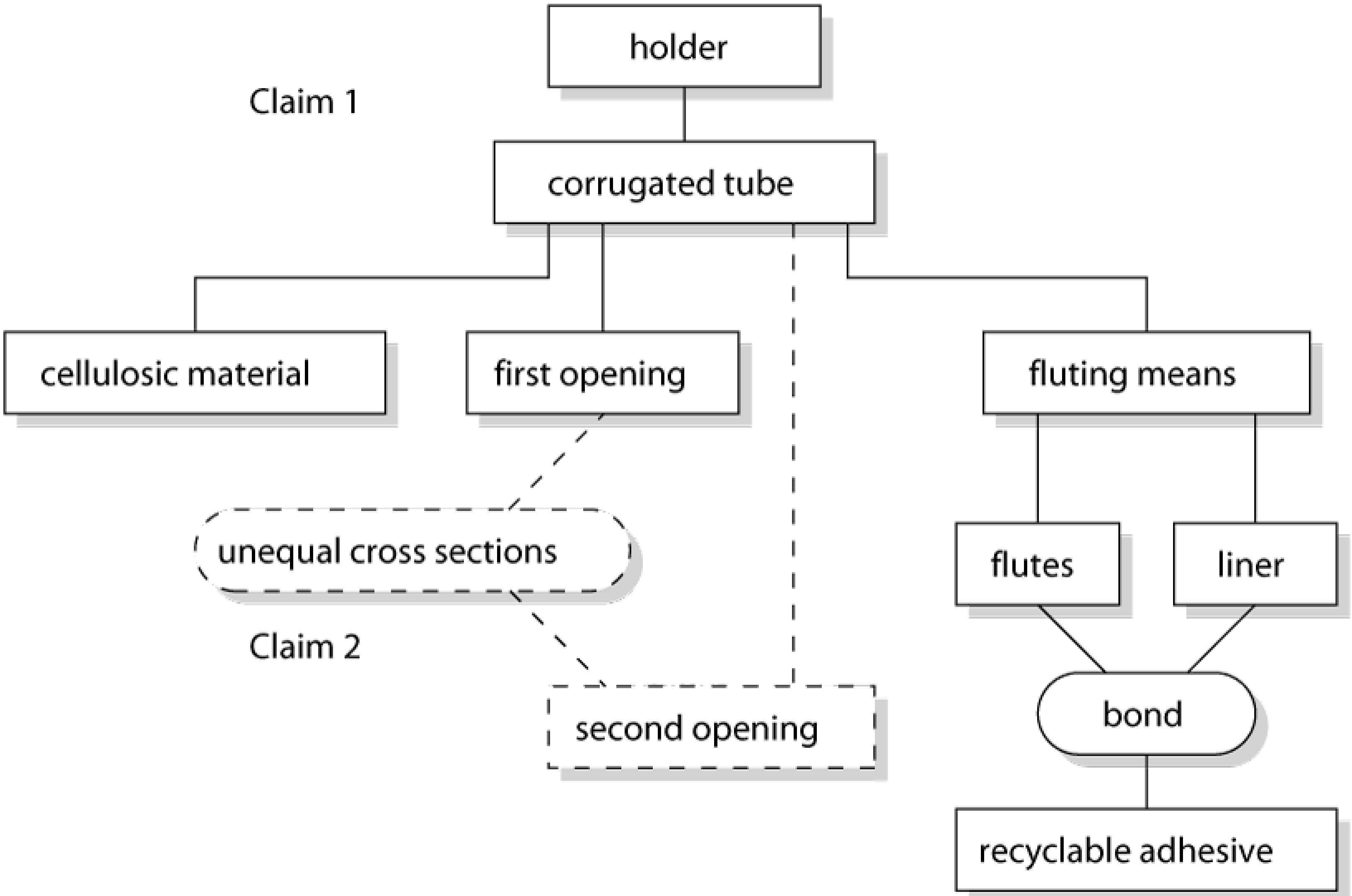
[57] **ABSTRACT**

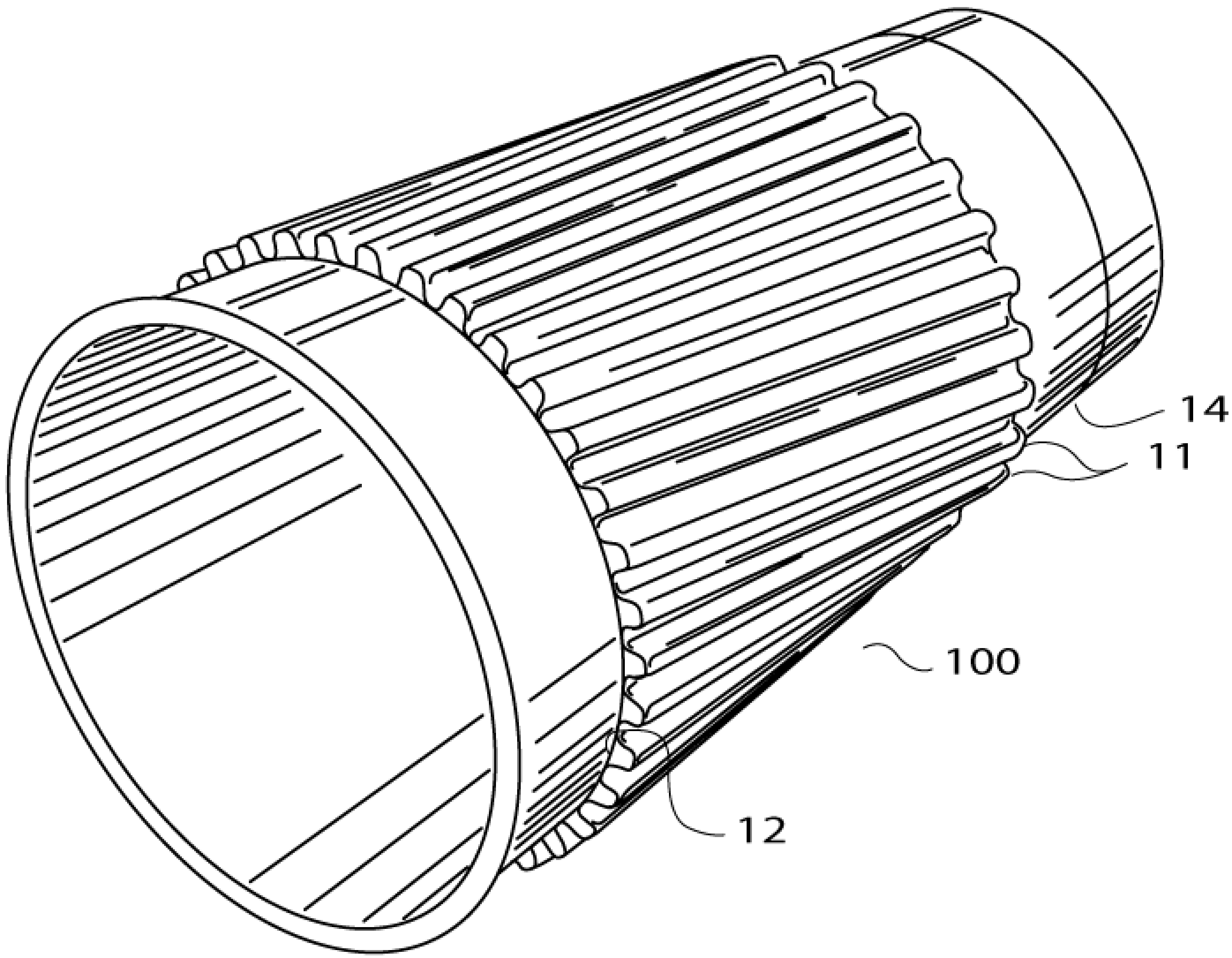

Corrugated beverage containers and holders are which employ recyclable materials, but provide fluting structures for containing insulating air. These products are easy to hold and have a lesser impact on the environment than polystyrene containers.

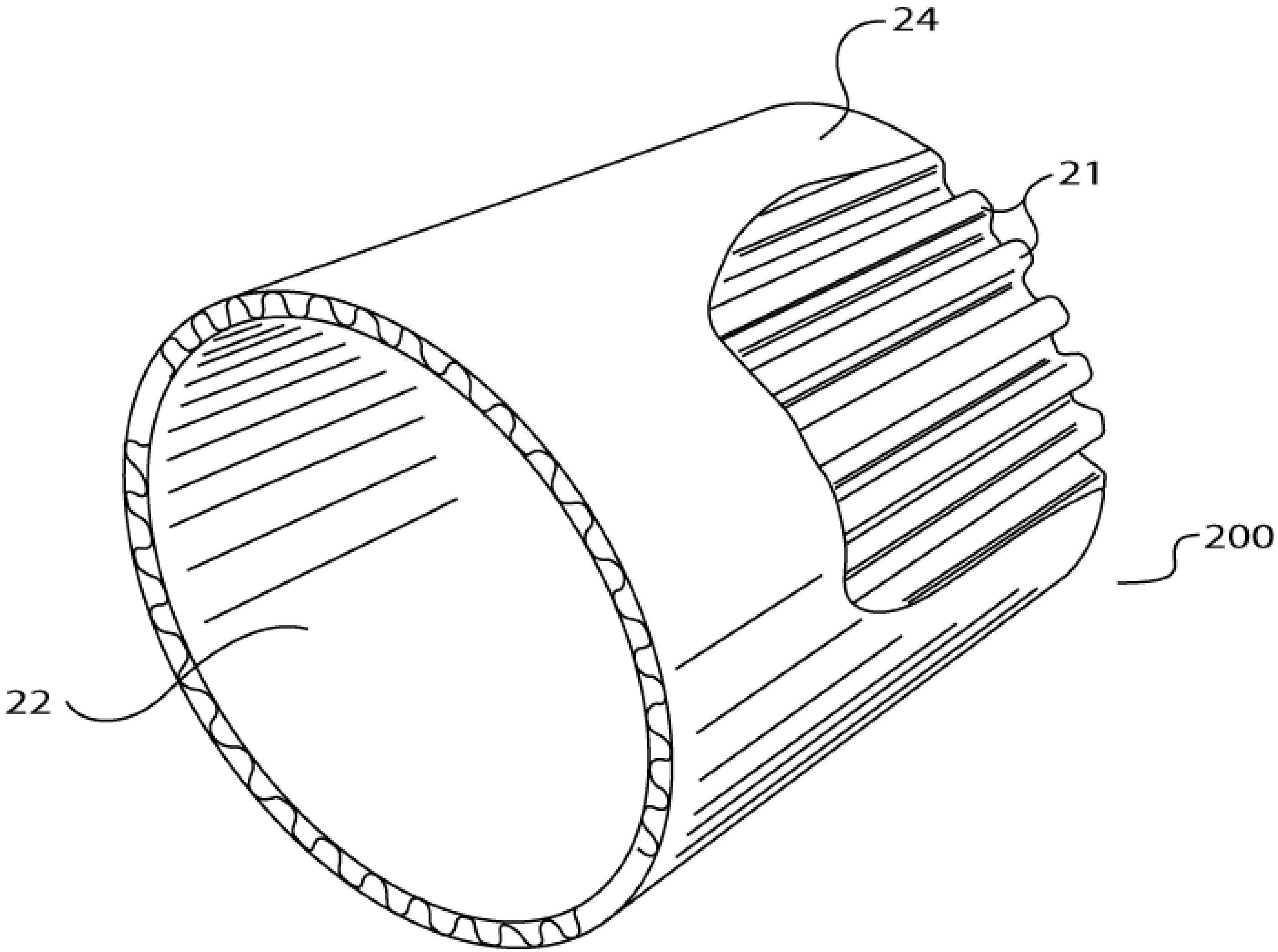

**18 Claims, 8 Drawing Sheets**



# D.W. Coffin Sr., US Pat.# 5,205,473: Claims 1 and 2




14
11
100
12


24
21
200
22

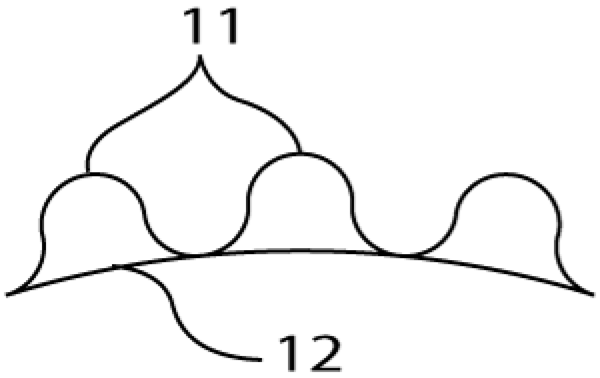



Fig. 6a

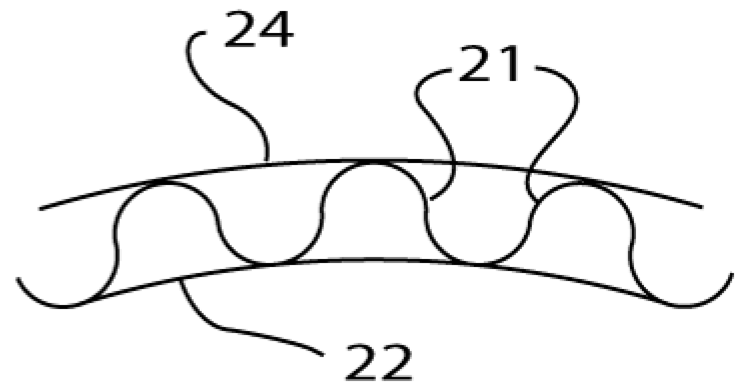



Fig. 6b

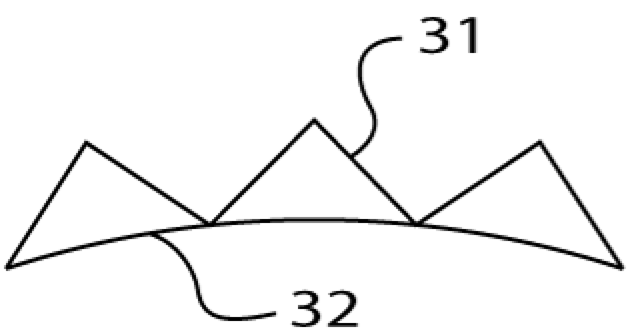



Fig. 7a

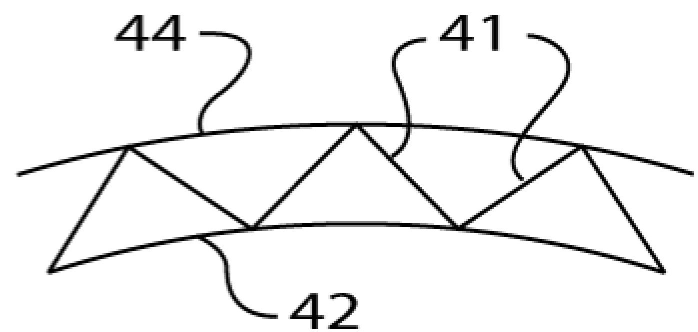
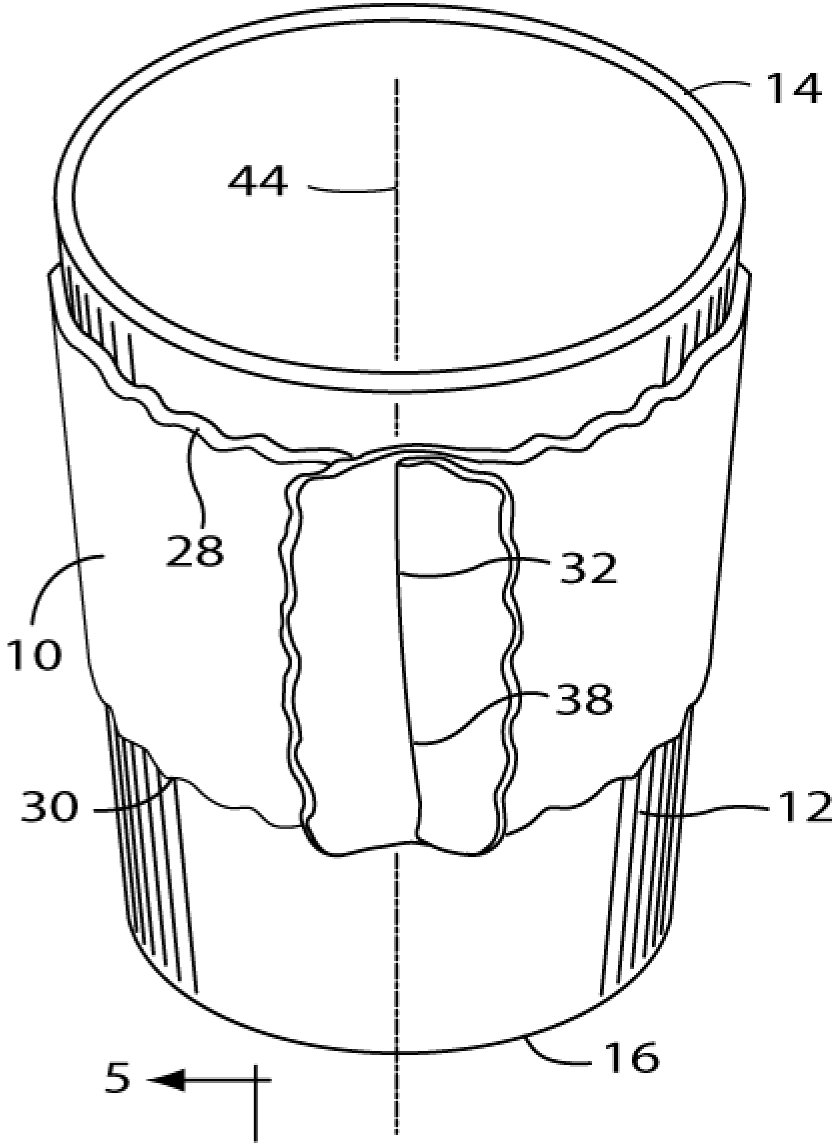
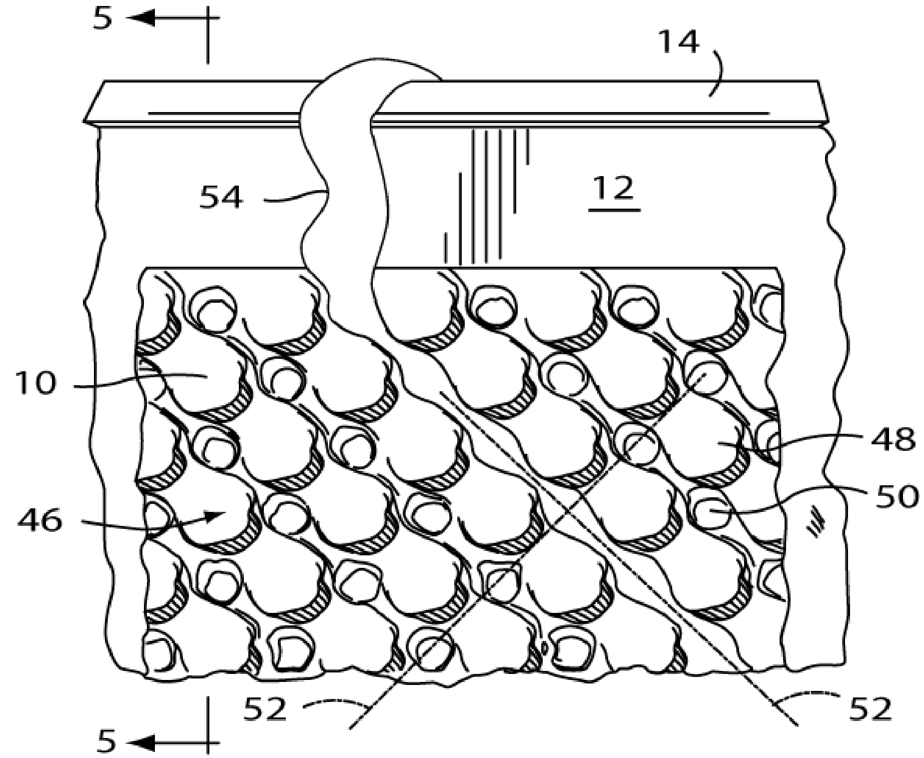



Fig. 7b

Jay Sorenson,
US Pat. # 5,425,497

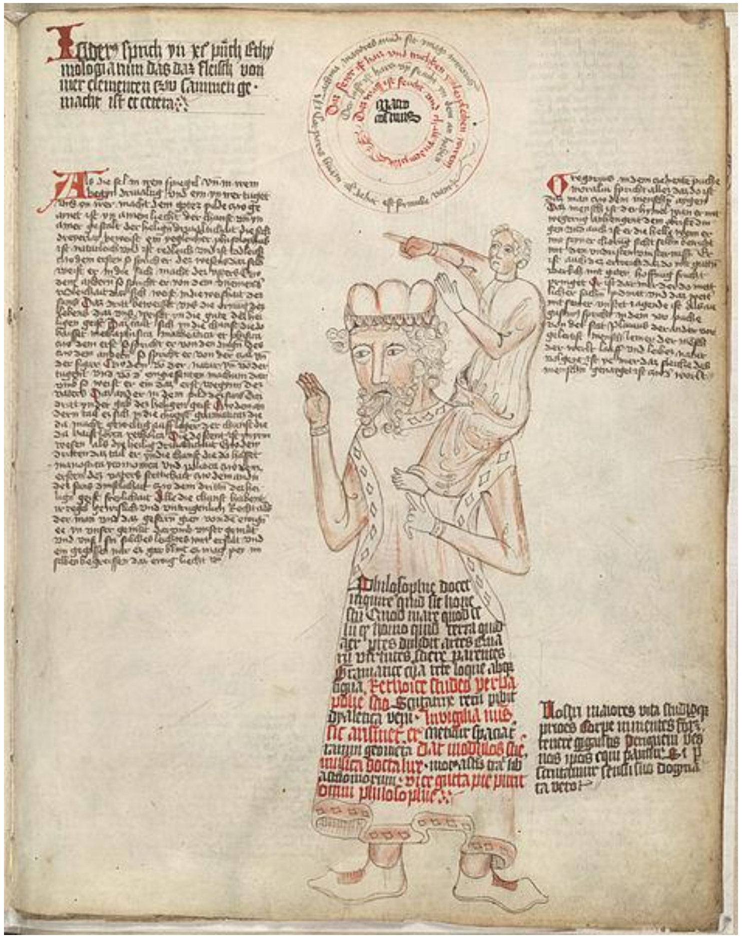


# Strategic Tools for Intellectual Property

- Publication
  - Cheapest way to prevent a competitor from patenting
- Patent
  - Gives limited right to exclude others
- License
  - Gives limited right to use patented invention under specified terms
- Non-Disclosure Agreement (NDA)
  - Prohibits disclosure or use of confidential information
  - Helps to protect patentability and trade secrecy
- Non-Compete Agreement (NCA)
  - Prevents employee from working for competitor (typically 3-36 months)
  - Not legally enforceable everywhere
- Employee Assignment (Invention) Agreement
  - Agreement to assign ownership of inventions while employed
- Material Transfer Agreement (MTA)
  - Controls how proprietary materials can be used
  - May provide for joint ownership of derived inventions
- Joint Invention Agreement
  - Specifies how to share costs, revenues, and IP responsibilities

# Standing on the shoulders of giants



Its most familiar expression is found in the letters of Isaac Newton:

“If I have seen further, it is by standing on the shoulders of giants.”

Your engineering or scientific development rely on other’s previous achievements.

But you should NOT use them without proper citation.

# Plagiarism

- “The act of passing off the information, ideas, or words of another as your own, by failing to acknowledge their source”
- “Means “kidnapper” in Latin” (“in antiquity, plagiarii were pirates who sometimes stole children.”)
- “When you plagiarize, you steal the brainchild of another.”

# In case of research article/paper

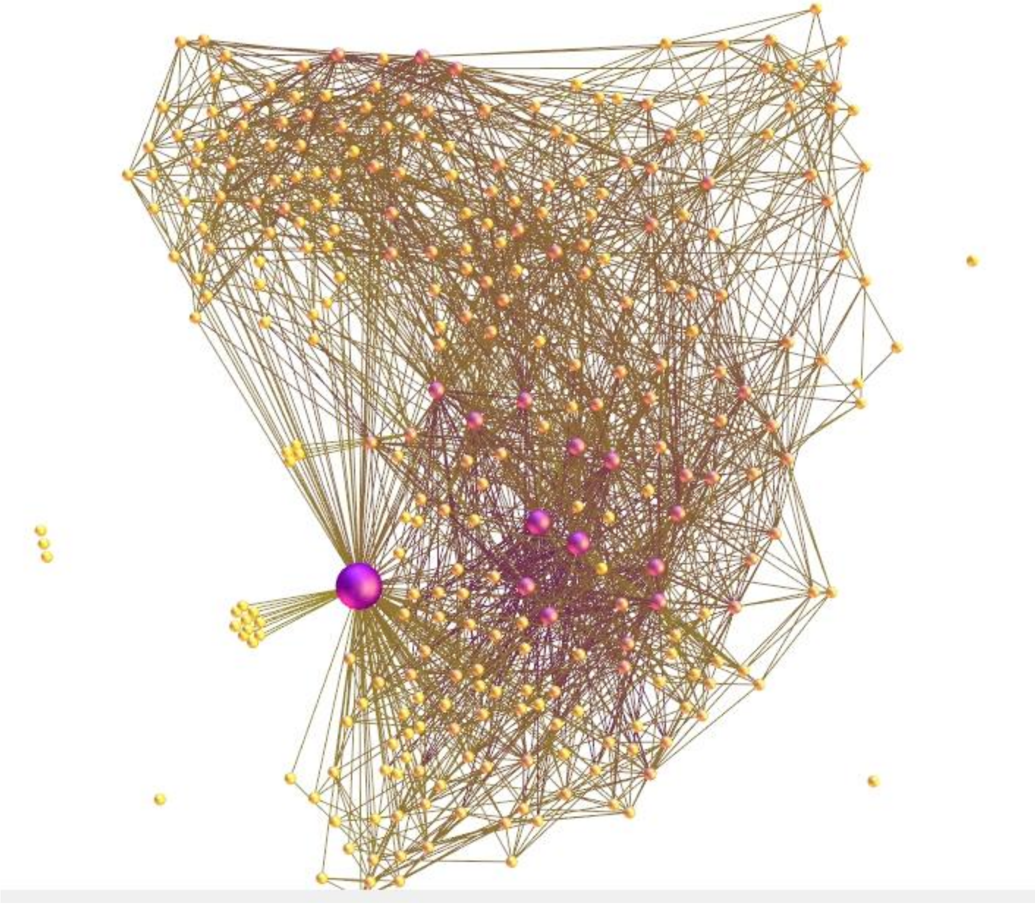
1. Self-plagiarism of text
2. Plagiarism of text from other authors
3. Self-plagiarism of results
4. Dual publication (a severe example of 3)
5. Representing others as co-authors who have not agreed
6. Plagiarism of other results and ideas
7. Scientific fraud and/or fabrication of data

More serious

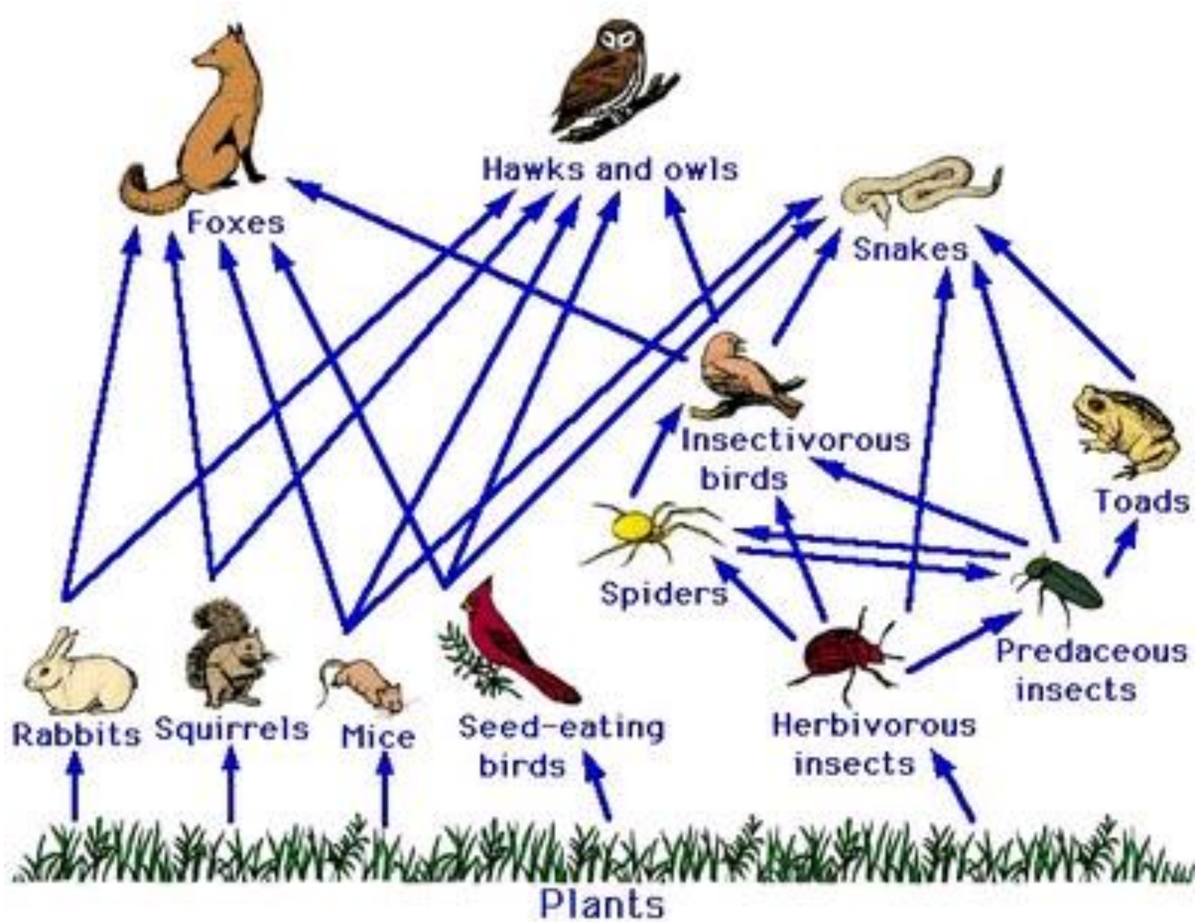
# Small World Networks

Engineering Idea #5

# Neural Network – C. Elegans Brain



# Ecological Network - Food Web



# Biological Network - Metabolism

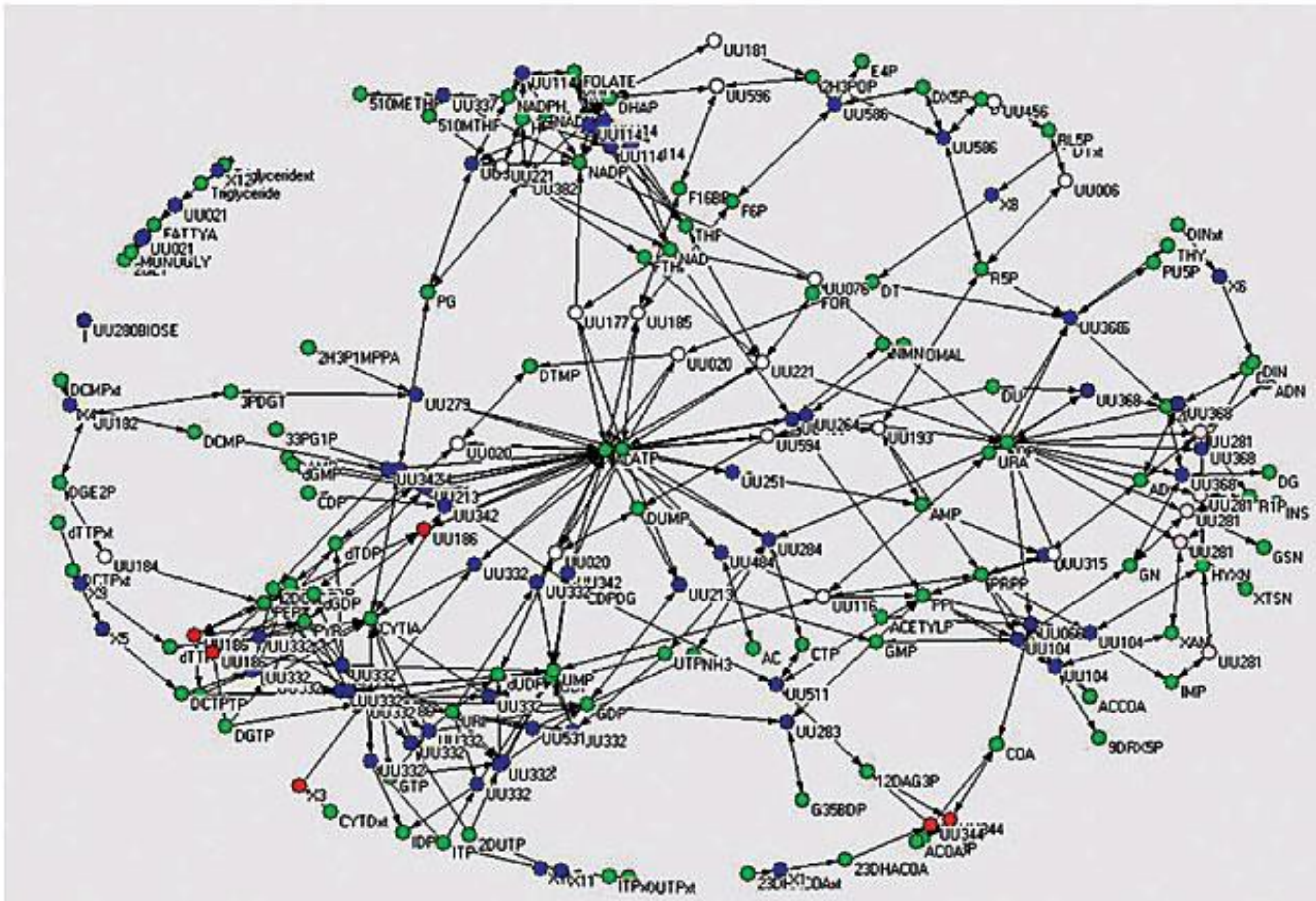
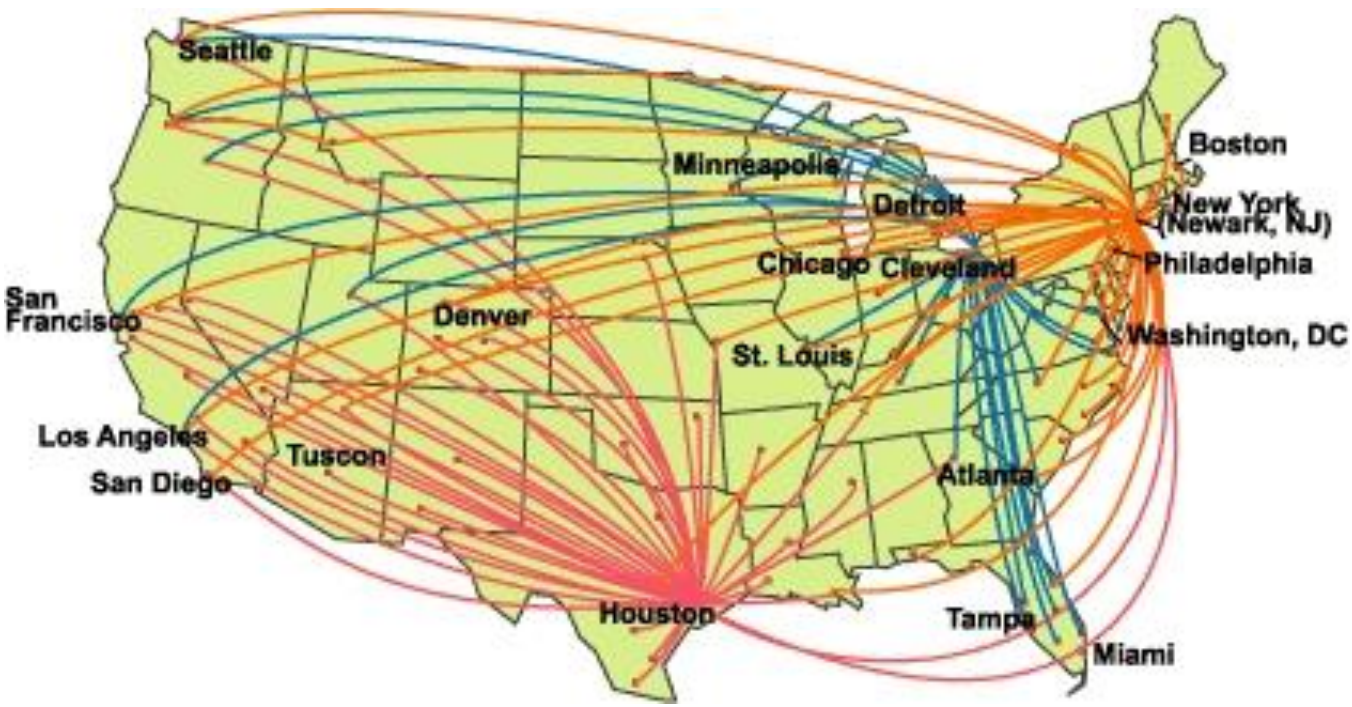
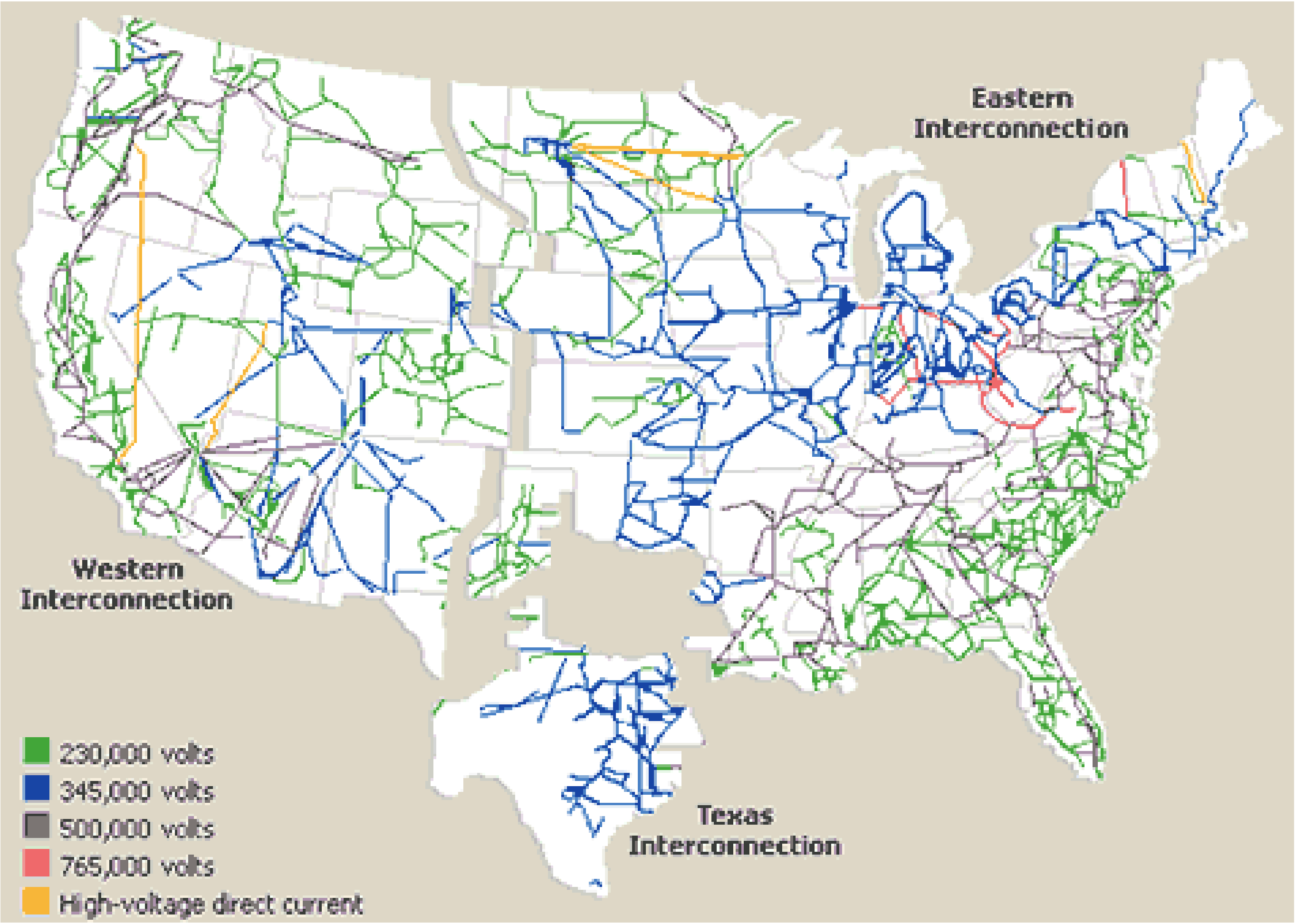



**Figure 2.** Bipartite graph of the metabolic network of *Ureaplasma urealyticum*. Dark gray and white nodes represent enzymes and light gray nodes represent metabolites (Lemke et al., 2004).

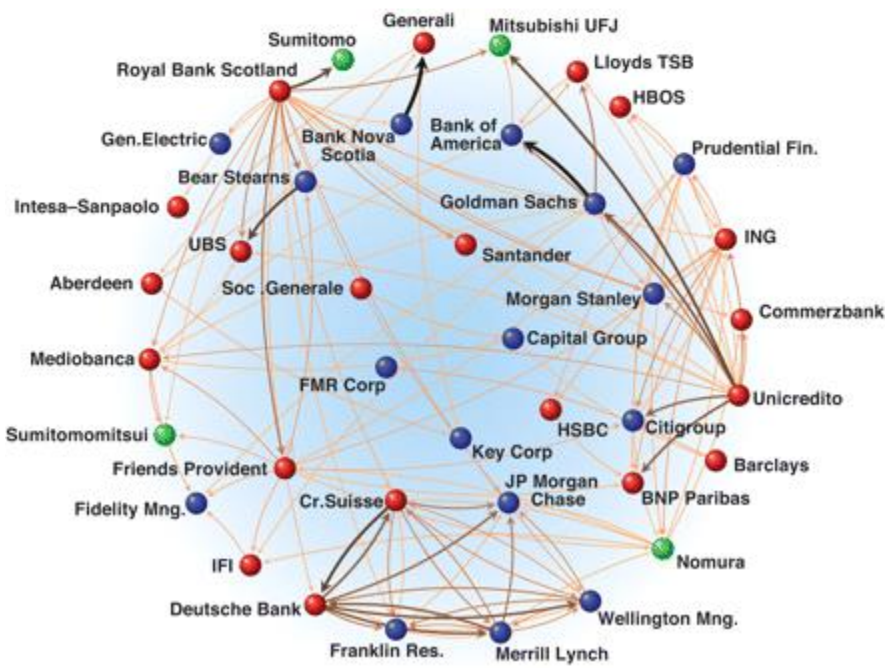
# Transport Network – Airline Routes



# Energy Network – Power Grid

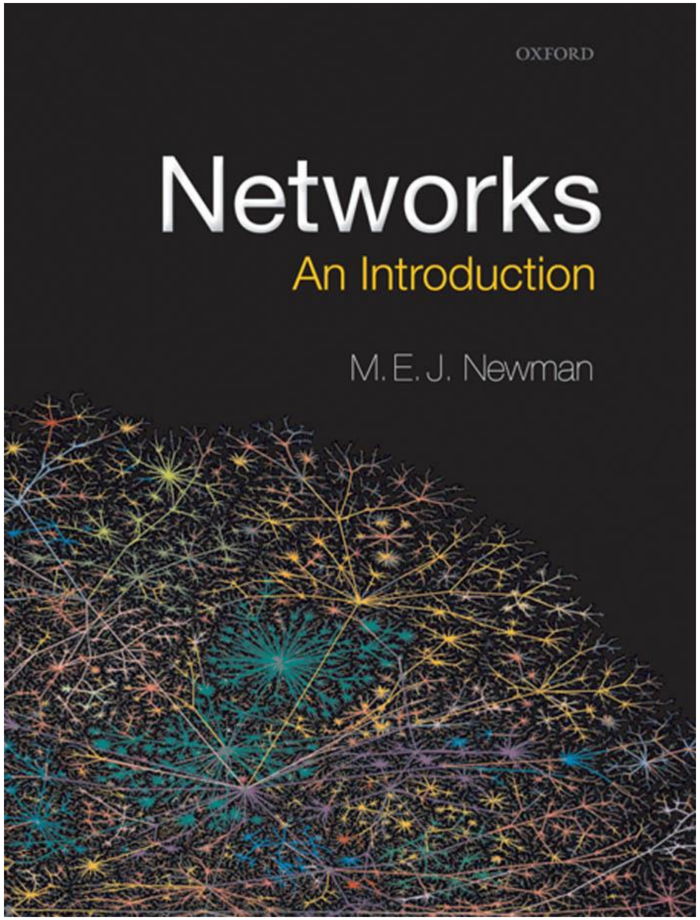


# Financial Network – Banks and their mutual liabilities

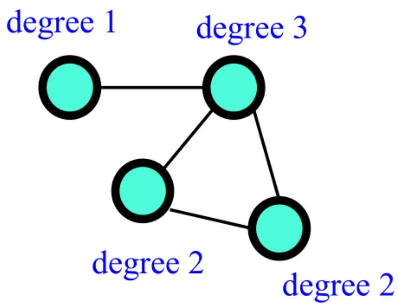


# Science of Networks

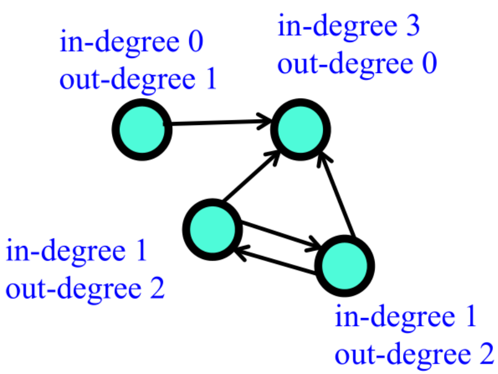
- Are there properties common to all of these networks? If so why?
- Can we formulate a general theory of the **structure**, **evolution** and **dynamics** of networks?
- Some common aspects of networks:
  - Small world property
  - Long-tailed degree distribution
  - Scale-free structure
  - Clustering and community structure
  - Robustness to random node failure
  - Vulnerability to targeted hub attacks
  - Vulnerability to cascading failures



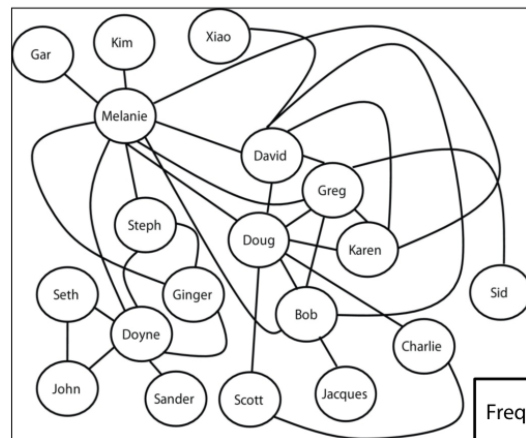
# Network Characteristics



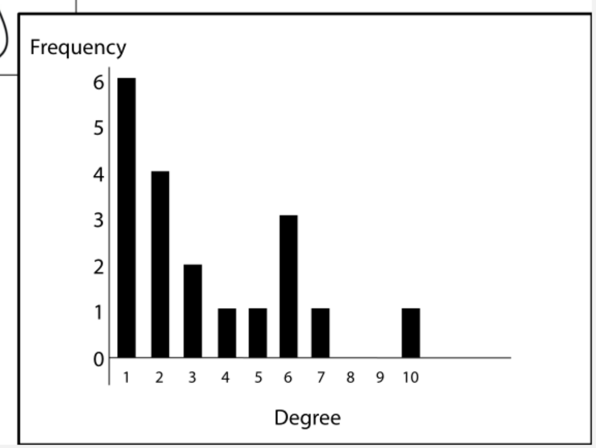

Undirected links and node degrees




Directed links and in/out degrees

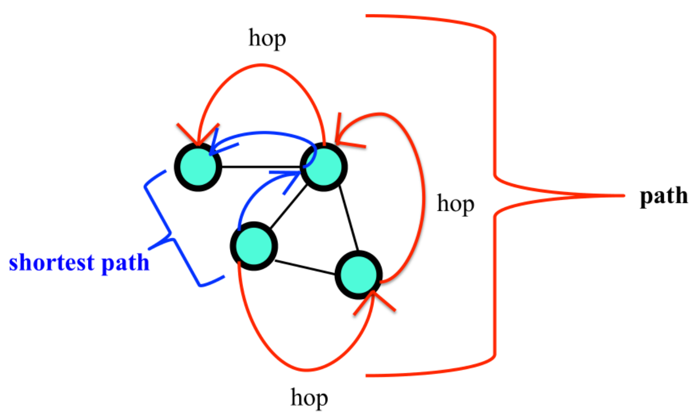



From M. Mitchell,
*Complexity: A Guided Tour*

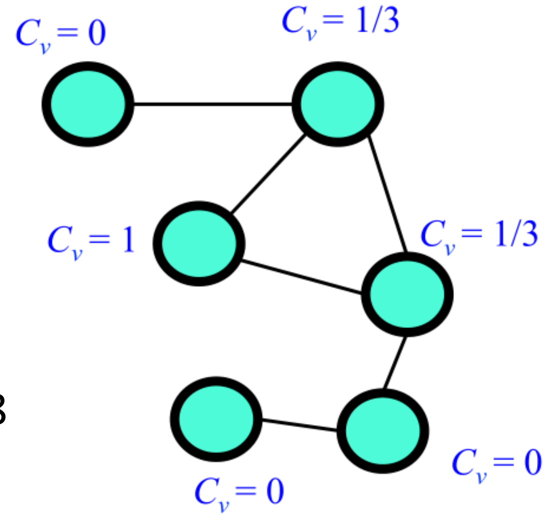


# Network Characteristics – Path and Clustering

- Distance between nodes is equal to the number of hop you should make

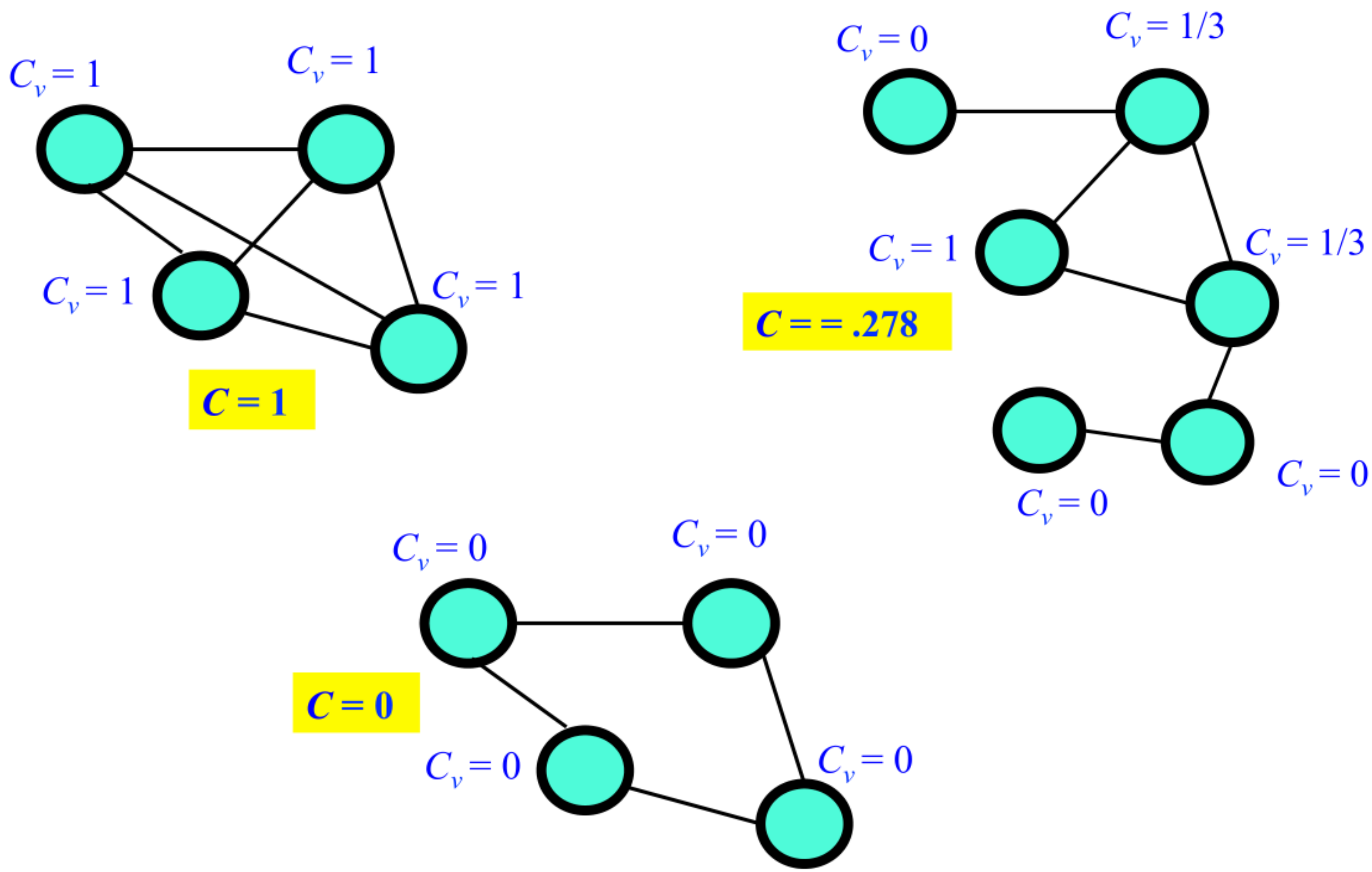



- **Clustering** – to what extent are your friends also friends of one another.
- Clustering of node *t*:
- $C_t = 2\frac{|\{e_{ij}:v_i,v_j \in N_t, e_{ij} \in E\}|}{k_t(k_t-1)}$

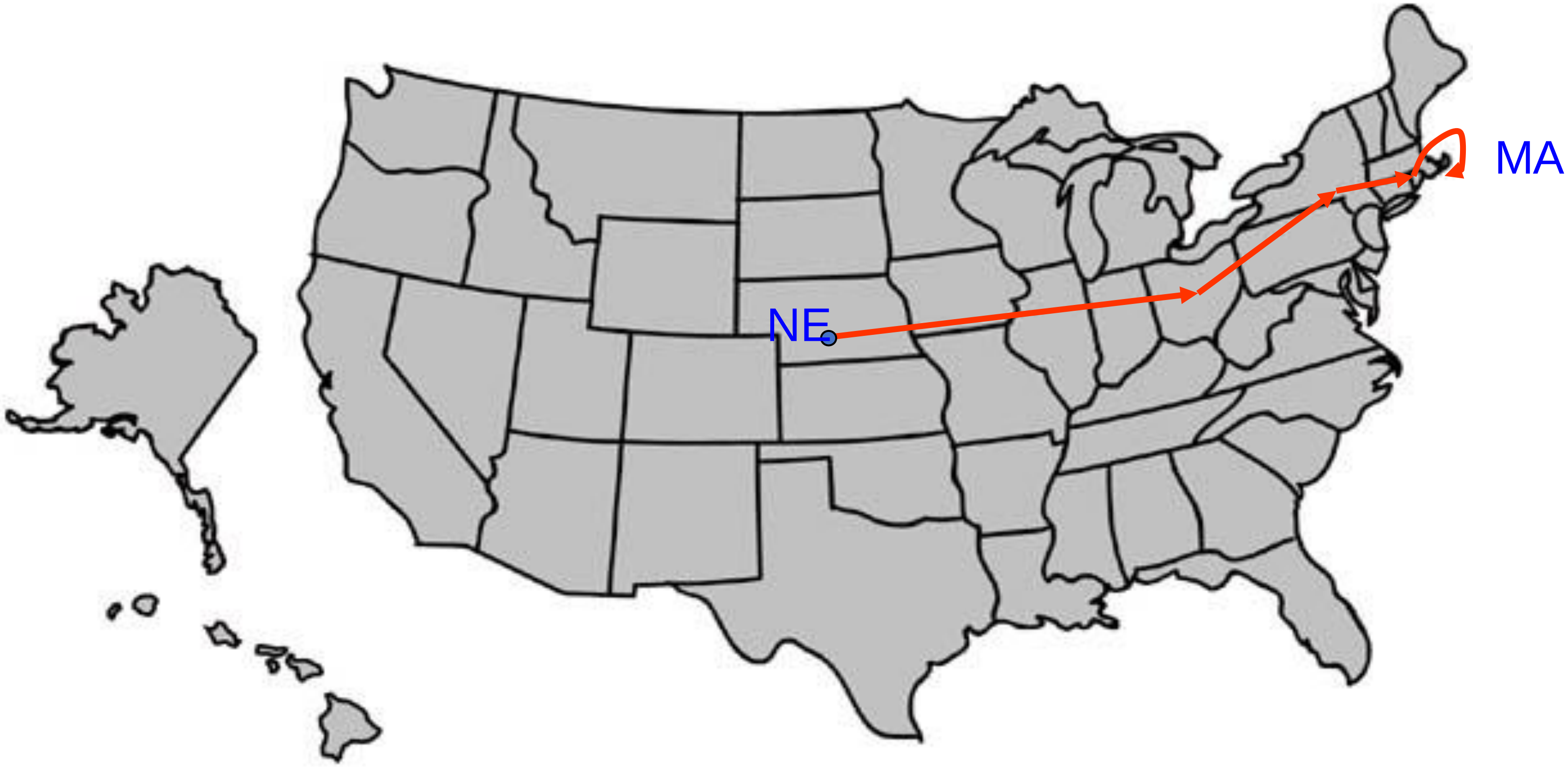



C = Average over all nodes. C = 1.67 / 6 = .278

# Clustering in Networks

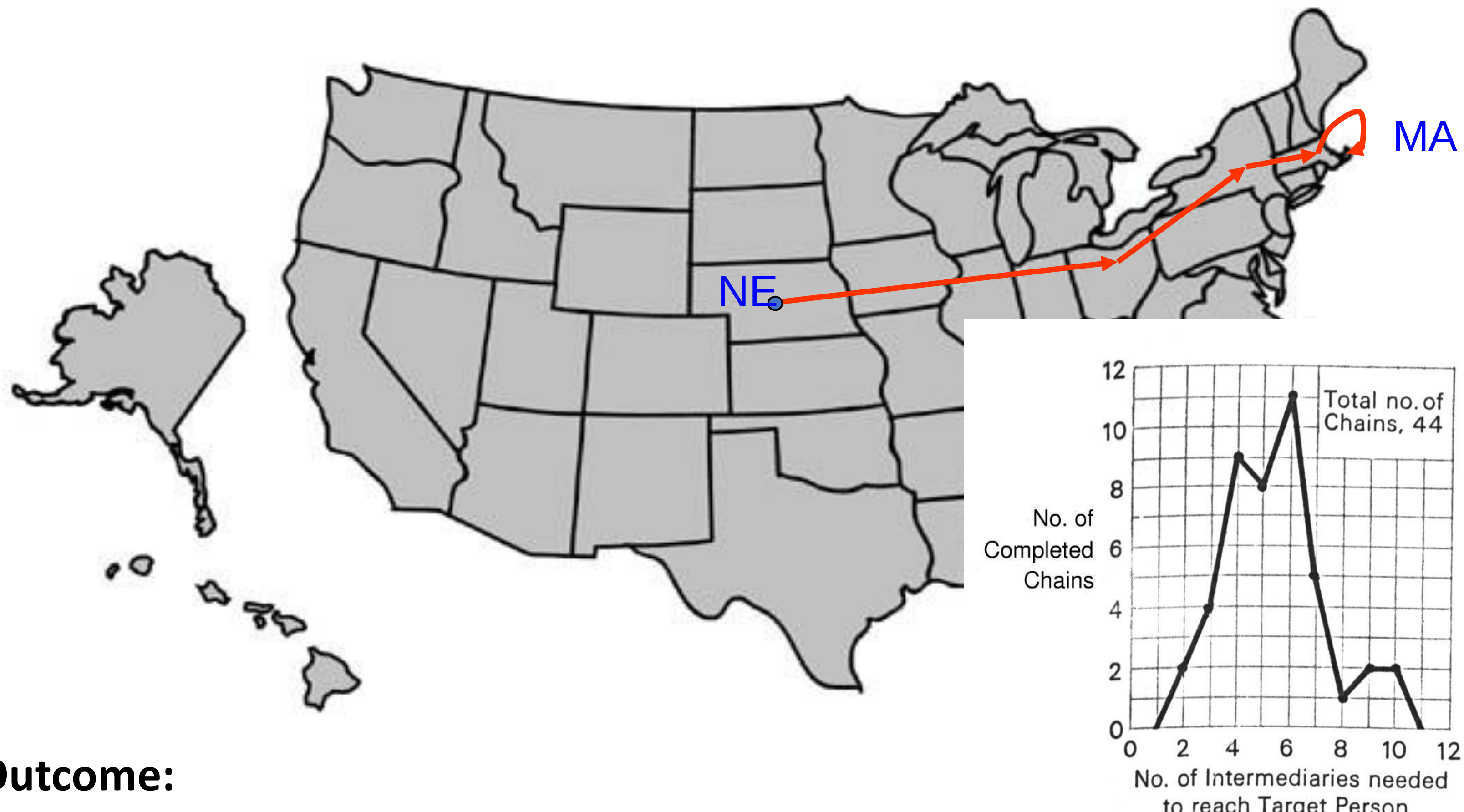


# Milgram’s experiment




**Instructions:**
Given a target individual (stockbroker in Boston), pass the message to a person you correspond with who is “closest” to the target.

# Milgram's experiment




**In the Nebraska Study the chains varied from two to 10 intermediate acquaintances with the median at five.**

**Outcome:**
**20% of initiated chains reached target**
**average chain length = 6.5**

"Six degrees of separation"

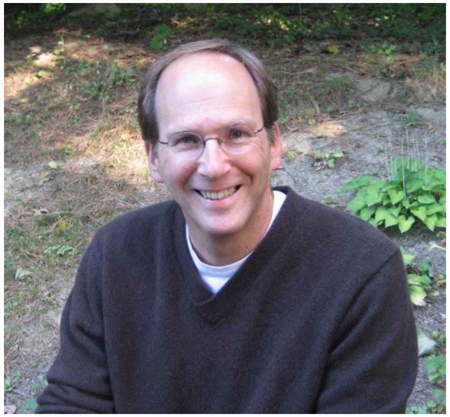
# Similar experiment with Emails

**Email** experiment
Dodds, Muhamad, Watts, Science 301, (2003)

- 18 targets
- 13 different countries

- 60,000+ participants
- 24,163 message chains
- 384 reached their targets
- average path length 4.0

# Real World Networks

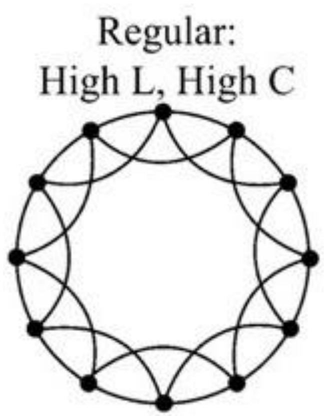


Steven Strogatz

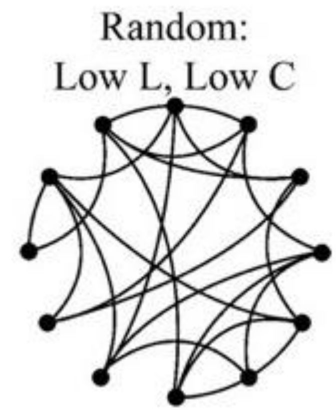
Duncan Watts

- Film actors:
  - Nodes = individual actors;
  - Two nodes are linked if the two actors appeared in the same movie.
- Power grid:
  - Nodes = generators, transformers, and substations;
  - Two nodes are linked if they have a transmission line between them.
- C. elegans:
  - Nodes = neurons;
  - Two nodes are linked if they are connected by a synapse or a gap junction

# Network Structure – Strogatz and Watts Model




High average distance
between nodes,
high average clustering



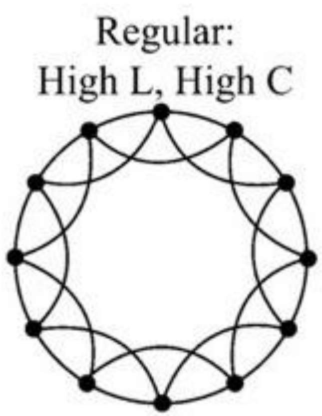

Low average distance
between nodes,
Low average clustering

# Strogatz and Watts' Empirical Results

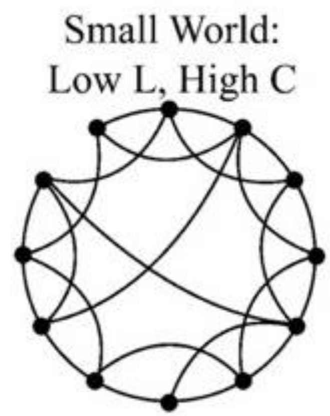
- $L_{actual}$ = Average distance (shortest path length) between pairs of nodes in the given network
- $L_{random}$ = Average distance (shortest path length) between pairs of nodes in a randomly connected network with the same number of nodes and links.
- $C_{actual}$ = Clustering coefficient of given network
- $C_{actual}$ = Clustering coefficient of random network

| | $L_{actual}$ | $L_{random}$ | $C_{actual}$ | $C_{random}$ |
|---|---|---|---|---|
| Film actors | 3.65 | 2.99 | 0.79 | 0.00027 |
| Power grid | 18.7 | 12.4 | 0.080 | 0.005 |
| *C. elegans* | 2.65 | 2.25 | 0.28 | 0.05 |

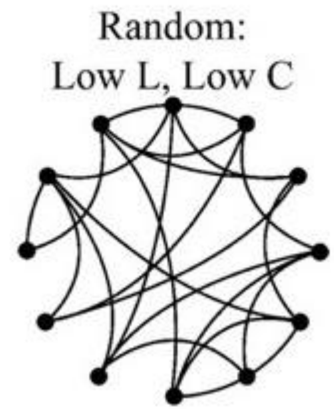
# Network Structure – Strogatz and Watts Model




High average distance
between nodes,
high average clustering



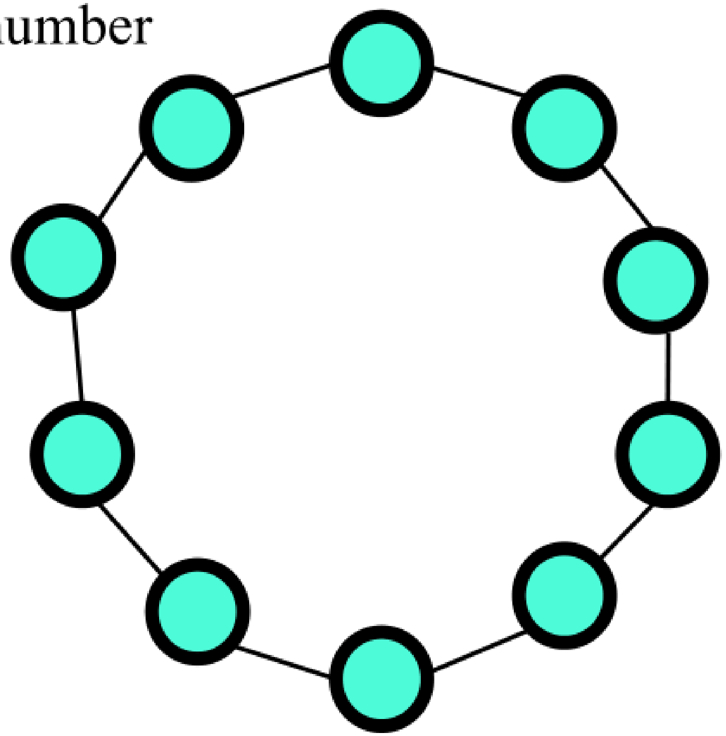

Low average distance
between nodes,
high average clustering




Low average distance
between nodes,
Low average clustering

# Network Structure – Strogatz and Watts Model

- Start with a *regular* network with $N$ nodes and $m$ neighbors per node. Let M be the number of links.



**For this example,**

$N = 10$

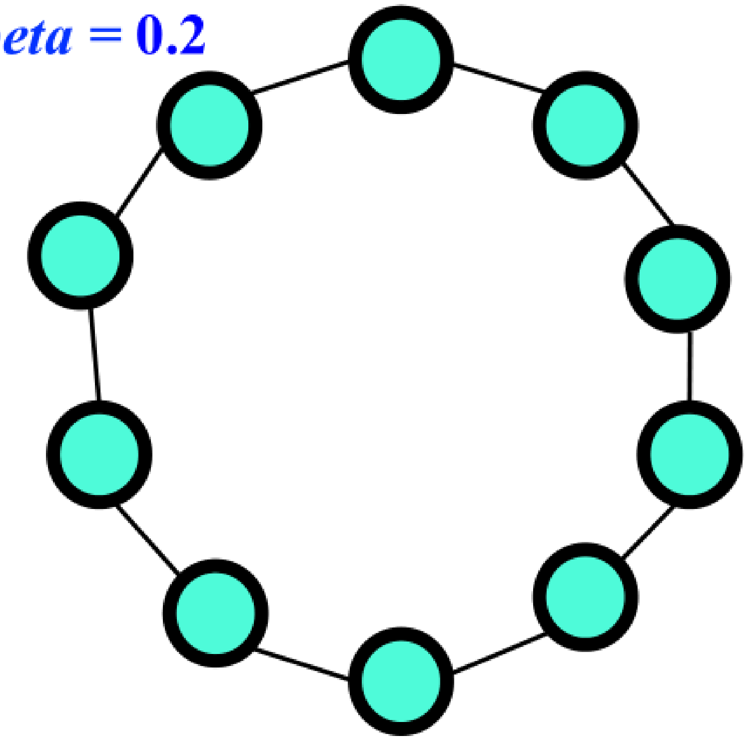
$m = 2$

$M = 10$

# Network Structure – Strogatz and Watts Model

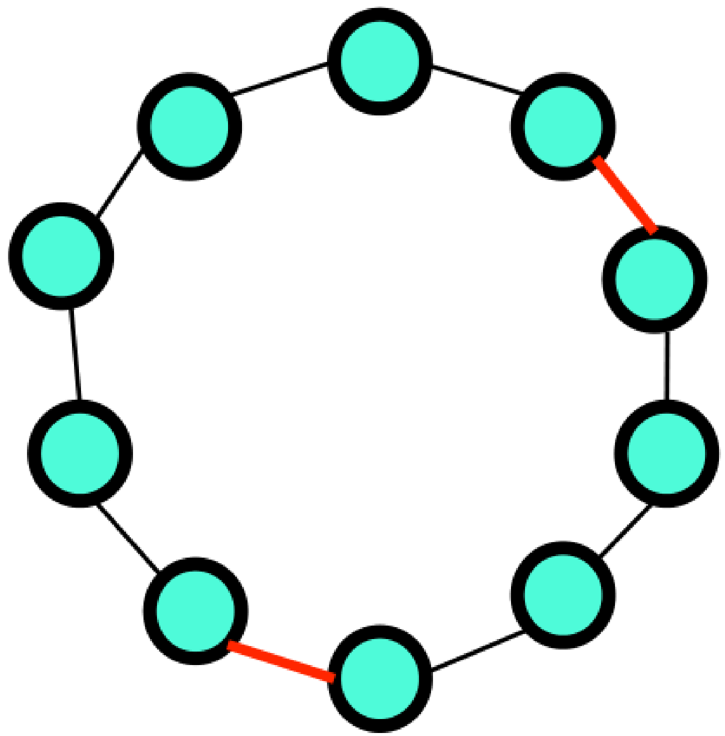
- Define *beta*, the probability of *rewiring*.

**For this example, let *beta* = 0.2**



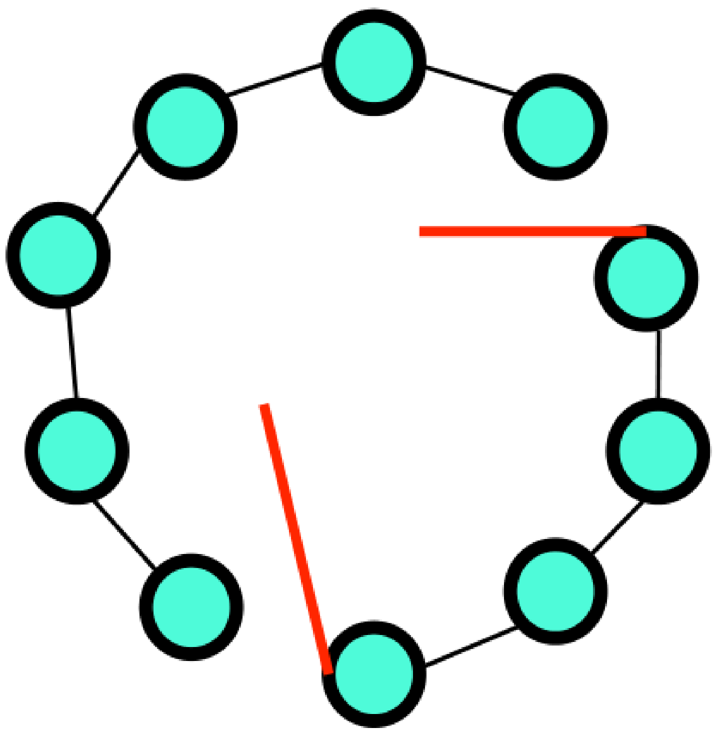
# Network Structure – Strogatz and Watts Model

Randomly choose *beta* × *M* links. **For this example, we choose 0.2 ×10 = 2 links.**



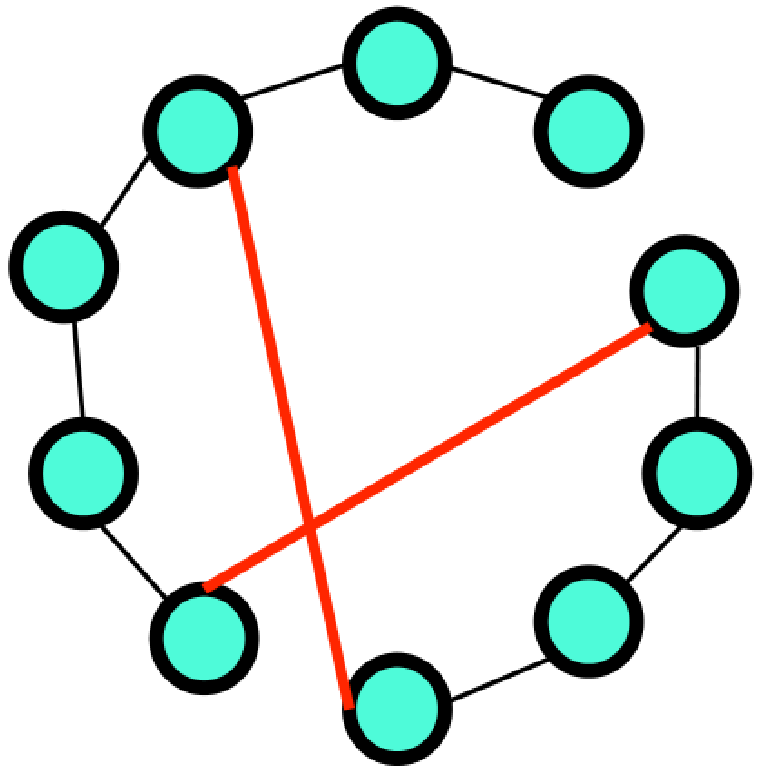
# Network Structure – Strogatz and Watts Model

Rewire one end of the chosen links to another randomly chosen node.

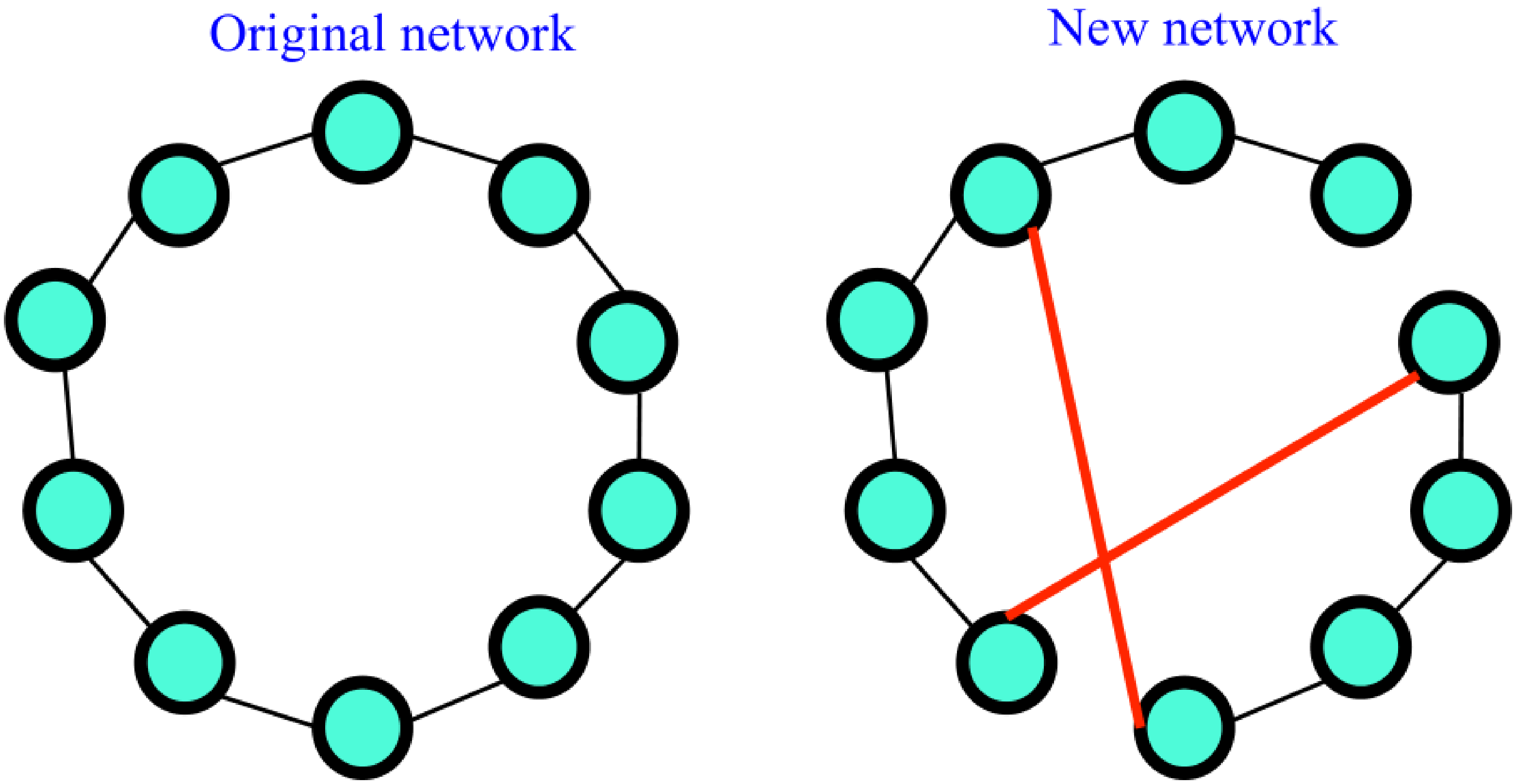


# Network Structure – Strogatz and Watts Model

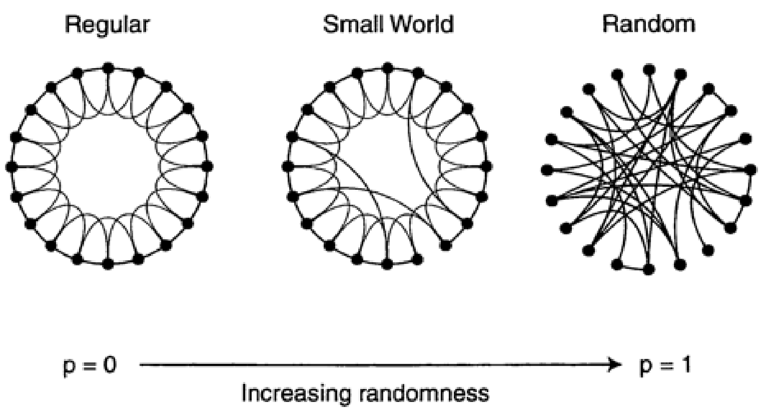
Rewire one end of the chosen links to another randomly chosen node.

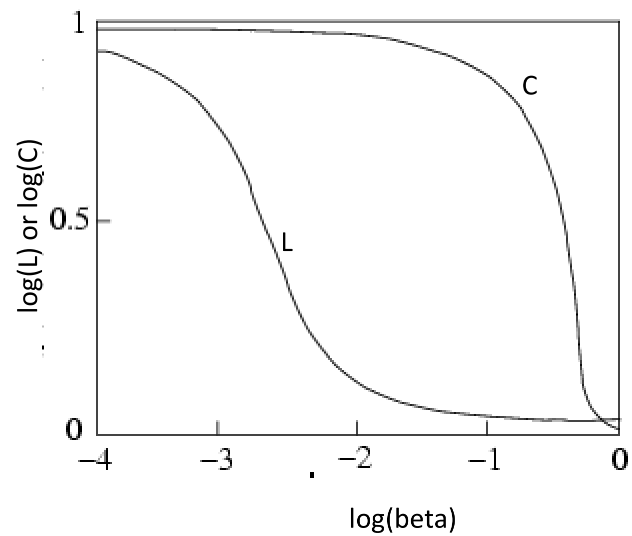


# How Small World Network is Formed



# Strogatz and Watts Model – Key Insight

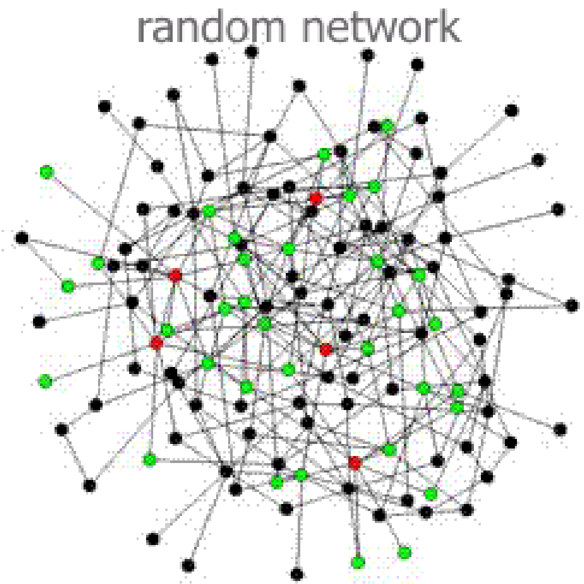


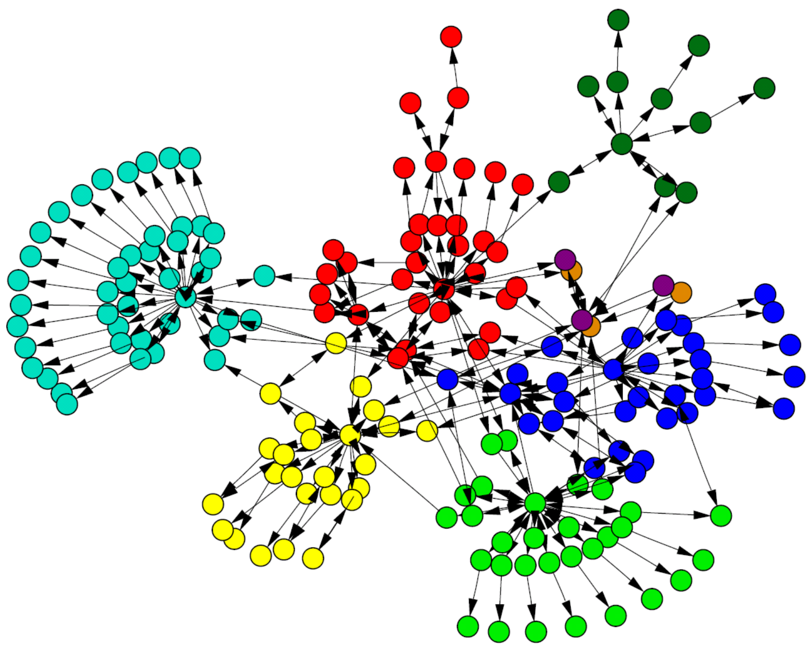

Where:
L – average length of shortest path between nodes
C – average clustering coefficient of nodes

# Preferential Attachment

- Random Network

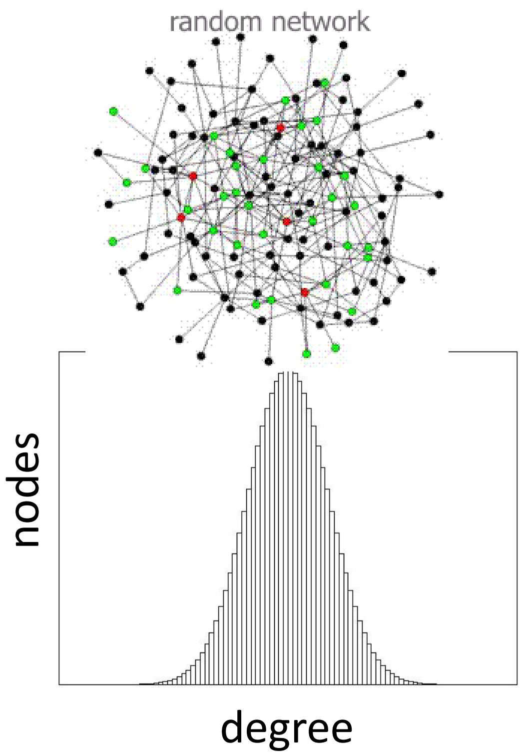



- Part of WWW

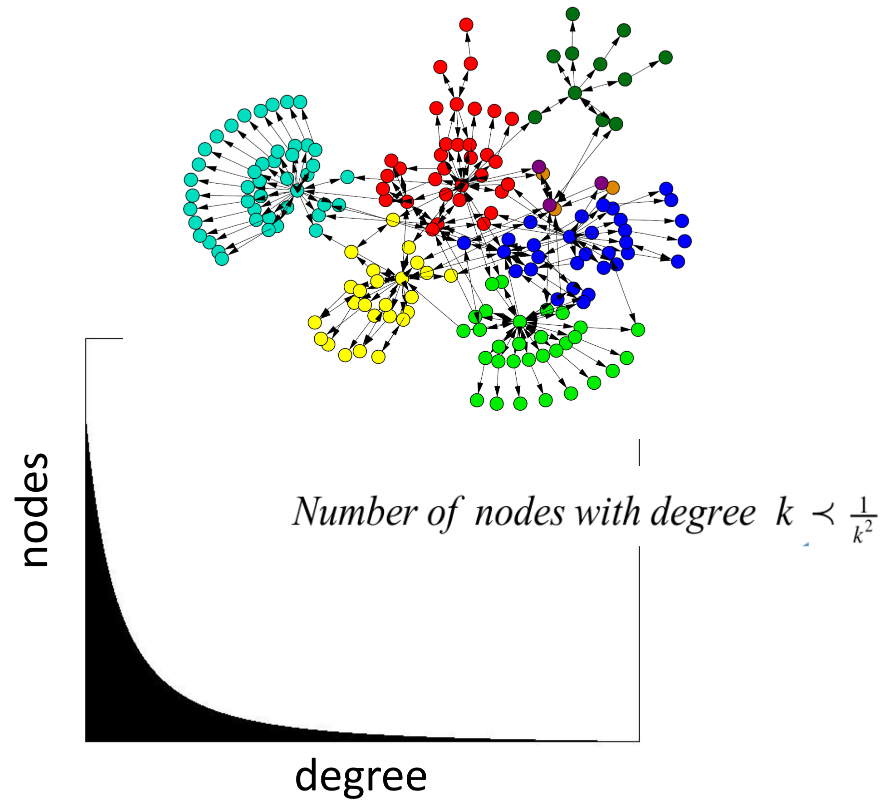


# Preferential Attachment

- Random Network




- Part of WWW



# References


- D. J. Watts and S. H. Strogatz, [Collective dynamics of 'small-world' networks].
- A. L. Barabasi and R. Albert, [Emergence of scaling in random networks].
- M. E. J. Newman, [Networks: An Introduction].